

Bayesian model averaging and the conditional volatility process: an application to predicting aggregate equity returns by conditioning on economic variables

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This study revisits the topic of predicting aggregate equity returns out-of-sample by conditioning on economic variables through Bayesian model averaging (BMA). Besides simultaneously addressing parameter instability and model uncertainty, I suggest a new model feature, namely, predictors in a given model can also impact the dependent variable through the conditional volatility process. The suggested econometric framework is straightforward to implement without requiring simulation. Likewise, the user can easily decide, which aspects of the predictive channel should be switched on, off or altered. I apply the suggested framework to the well-known [Goyal, A. and Welch, I., A comprehensive look at the empirical performance of equity premium prediction. *Rev. Financial Stud.*, 2008, **21**, 1455–1508] dataset. An extensive out-of-sample prediction evaluation demonstrates that averaging over predictor combinations in a model that allows lagged predictors to impact aggregate equity returns exclusively through the conditional volatility process results in statistically significant more accurate density predictions relative to the benchmark, especially when predicting the left tail of the conditional distribution. One also observes economic gains in favor of certain BMAs. Here, the BMA that allows predictors to impact equity returns through the conditional mean as well as the conditional volatility process is the top performer.

Keywords: Aggregate equity return predictability; Bayesian model averaging; Conditional volatility; Density prediction accuracy

JEL Classification: C11, C22, C51, G17

1. Introduction

This study suggests a new Bayesian model averaging (BMA) framework with application to predicting aggregate equity returns by conditioning on economic variables.[†] The point of departure is from seminal studies, such as Campbell and Thompson (2008), Goyal and Welch (2008), Rapach *et al.* (2010), Dangl and Halling (2012), Rapach and Zhou (2013), among others.[‡] For example, using univariate predictive regressions (in the sense that each equity return is

regressed on a constant and the lagged economic variable of interest) estimated via OLS, Goyal and Welch (2008) were among the first to illustrate that evidence of out-of-sample predictability relative to the historical average benchmark is weak and prone to instability over time.[§] Conversely, studies such as Avramov (2002), Cremers (2002), Rapach *et al.* (2010), Dangl and Halling (2012), Rapach and Zhou (2013) find that there are notable out-of-sample

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[†] Clearly, the framework can be applied in other fields of economics, such as inflation and GDP growth rate forecasting.

[‡] Campbell and Thompson (2008) are the first to document that imposing ‘economic’ constraints on the sign of the regression coefficient of the predictive model and return prediction leads to a better out-of-sample performance relative to the benchmark.

[§] This is in contradiction to in-sample based results, see for example, Campbell and Shiller (1988), Stambaugh (1999), Ang and Bekaert (2007), Lettau and Nieuwerburgh (2008), Pastor and Stambaugh (2009). A number of studies conclude that this contradiction is because in-sample analysis tends to suffer from data mining, see Lo and MacKinlay (1990), Foster *et al.* (1997) for details. Furthermore, at the time, results reported in Goyal and Welch (2008) caused a huge controversy and debate in the literature. Consequently, the *Review of Financial Studies* (Vol. 21, No.4, 2008) dedicated a special issue to this topic in 2008.

point prediction accuracy gains to be made by combining individual return predictions. Using the dynamic linear model framework described extensively in West and Harrison (1997), Dangl and Halling (2012) suggest an econometric framework that deals simultaneously with parameter instability and model uncertainty through BMA.[†] Using the Goyal and Welch (2008) data from 1937 to 2002, they find very interesting results: First, they demonstrate strong out-of-sample evidence in favor of their time-varying regression coefficient BMA relative to constant-parameter BMA, univariate alternatives and the historical average benchmark. Second, even setting aside model uncertainty, univariate time-varying regression coefficient models tend to dominate their time-invariant counterparts, highlighting the importance of accounting for this feature at the modeling stage. Third, an investor who relies on point predictions produced under the suggested BMA tends to time the market very well, reducing her exposure around the peak of the business cycle and moving back into the market before the trough.

Finally, studies such as Campbell and Thompson (2008), Pettenuzzo *et al.* (2014), Pan *et al.* (2020) find that imposing economic restrictions on the model leads to (population-level) predictability gains relative to the historical average benchmark. More explicitly, using univariate predictive regressions, Campbell and Thompson (2008) impose that (a): The loading coefficient on the lagged economic variable of interest must be of the sign assumed by economic theory, otherwise prediction from the historical average benchmark is used, (b): The return prediction must be positive, otherwise the point prediction is set equal to zero and (c): (a) and (b) combined. Using a similar framework as Campbell and Thompson (2008), Pettenuzzo *et al.* (2014) suggest a Gibbs sampling methodology for imposing constraints that rule out negative returns as well as bounding the conditional Sharpe ratio. Rather than (a)–(c) or bounding the Sharpe ratio, Pan *et al.* (2020) impose the restriction on the predictor itself, namely, setting it to zero whenever its value falls within the predictor's rolling 24-month high and low. They find that a simple average combination of predictions under the suggested framework outperforms those obtained using techniques outlined in Campbell and Thompson (2008), Pettenuzzo *et al.* (2014).

Regardless of whether researchers account for parameter instability, model uncertainty or even specify economic constraints on the model of interest, almost all studies in the current literature have one major feature in common: First, the predictors of interest are constructed. For example, the divided yield is constructed as the difference between the logarithm of dividends on the S&P 500 index and logarithm of

the S&P 500 index lagged one month. Second, the lagged variables enter into the model as regressors to have their predictive power evaluated using the user-specified loss function(s). However, to my knowledge, no attempts have thus far been made in equity return predictability nor the broader forecasting literature to evaluate whether there are more predictive gains to be made by either allowing predictors in the BMA to impact the dependent variable exclusively through the conditional volatility process or through the conditional mean as well as the conditional volatility process.[‡]

The purpose of this study is to suggest a BMA framework that can accommodate (i): Model uncertainty, (ii): Parameter instability and a new feature, namely, (iii): Allow the predictors to impact the dependent variable through the conditional volatility process. I deal with (i) and (ii) by recursively averaging over the total possible number of predictor combinations as well as degrees of instability in the regression coefficients.[§] With regards to incorporating (iii), I follow (Koop and Korobilis 2014) and use dual linear Kalman filters, see Section 2.1 for details. The idea to incorporate (iii) is inspired by the recent study of Maheu *et al.* (2020). The authors suggest univariate predictive regressions, where besides the usual conditional mean channel, the lagged value of the predictor of interest can also impact the dependent variable through the conditional volatility process. Their application focuses on forecasting the U.S. real GDP growth rate by conditioning on various nonlinear transformations of the price of crude oil using univariate models widely applied in the literature.[¶] As illustrated in Maheu *et al.* (2020), the specification that accounts for the predictive impact only through the conditional volatility process leads to density forecast accuracy gains relative to the competitors and the benchmark at longer forecast horizons. Recently, Nonejad (2020) applies a univariate predictive regression similar to the one suggested in Maheu *et al.* (2020) to predict aggregate equity returns out-of-sample by conditioning on different economic variables one by one. He finds that the specification that accounts for the predictive impact only through the conditional volatility process affords statistically significant density prediction accuracy gains relative to the benchmark. The observed levels of statistical gains can also be translated into economically meaningful gains using the Value-at-Risk (VaR) framework.

[†] There are several reasons for why model parameters might vary over time. Examples include changes in regulatory conditions, market sentiments and macroeconomic policy. The same goes for model uncertainty since predictors that can help explain variation in the dependent variable at certain time-periods might not do so in other time-periods. I refer the reader to studies, such as Viceira (1997), Pastor and Stambaugh (2001), Kim *et al.* (2005), Paye and Timmermann (2006), Johannes *et al.* (2014), where the authors focus on modeling structural breaks/time-variation in the model parameters using Markov-switching specifications or models that assume gradual changes in the regression coefficients. Finally, Avramov (2002), Cremers (2002) are the first studies that apply BMA to equity return prediction.

[‡] It should be mentioned that in the financial literature there are studies that use various model combination schemes, such as BMA to forecast equity return volatility, see for example, Christiansen *et al.* (2012), Paye (2012), Nonejad (2017). These studies mainly use an ex-post measure of equity market volatility by computing realized volatility at the monthly sampling frequency using daily data. This approach in turn, implies that volatility is observed and one can perform BMA by considering monthly realized volatility as the dependent variable and lagged volatilities as well as lagged economic variables, such as the dividend yield and the default return spread at the monthly sampling frequency as predictors. However, this approach cannot be easily extended to other applications, where higher frequency data is not available to construct realized volatility at a lower frequency. Examples include the rate of inflation and GDP growth rate.

[§] The Bayesian response to dealing with uncertainty is averaging, see Steel (2020) for an extensive survey on BMA.

[¶] I refer the reader to Kilian and Vigfusson (2013) for a detailed analysis of these nonlinear crude oil price measures.

The suggested framework in this study is more advanced than the univariate framework of Maheu *et al.* (2020). To begin with, I work in a multiple regression setting and simultaneously account for parameter instability as well as model uncertainty. Furthermore, contrary to the MCMC-based framework of Maheu *et al.* (2020), model estimation is conducted in real-time without the need resort to simulation nor to recursively re-estimate the model of interest as new observations become available. In a nutshell, the main contribution of this study is to develop a fast and straightforward econometric framework using dual Kalman filters that explicitly takes into account (i)–(iii), and show its usefulness with regards to an important topic in financial economics, namely, out-of-sample equity return prediction by conditioning on economic variables. From a technical viewpoint, I introduce a new BMA framework that can also be useful in other time-series applications. This is the computational contribution of the paper.

Following the bulk of studies focusing on predicting equity returns by conditioning on economic variables, I rely on the widely used (Goyal and Welch 2008) dataset, and perform an out-of-sample prediction evaluation based on monthly returns on the S&P 500 index as well as well-known financial ratios, such as the dividend yield, the default yield spread and other economic variables using data from 1926m12 through 2018m12.[†] The most interesting result is that in terms of density prediction accuracy, I observe consistent and statistically significant support for the BMA with time-invariant regression coefficients, where predictors impact aggregate equity returns exclusively through the conditional volatility process. For example, across almost all out-of-sample periods, it is the only BMA that generates statistically significant density prediction accuracy gains relative to the stochastic volatility benchmark when predicting the left tail of the conditional return distribution. I also observe that once the predictive impact is incorporated exclusively through the conditional volatility process, allowing for time-variation in the model parameters does not lead to any additional gains in density prediction accuracy worth mentioning. Likewise,

the time-varying regression coefficient BMA suggested in Dangl and Halling (2012), which assumes constant innovation volatility, fails to statistically outperform the stochastic volatility benchmark. In fact, results in this study indicate that the evidence in favor of time-varying regression coefficients BMA reported in Dangl and Halling (2012) is likely a consequence of neglecting time-varying volatility. Once this aspect is accounted for, allowing for time-varying regression coefficients does not lead to any sizable gains in density prediction accuracy. Lastly, simple predictor combination schemes and BMAs that impose the constraint suggested in Pan *et al.* (2020) on individual predictors tend not to afford gain in density prediction accuracy relative to the benchmark.

In the next step, I analyze the most successful models in more detail to understand why they perform better than the others. Again, several interesting results are observed. To begin with, the shrinkage of BMA depends heavily on which predictive channel is turned on (off). For the BMA with time-varying conditional innovation volatility, where predictors impact equity returns only through the conditional mean channel, the expected model size varies between 1 and 2 even though there are 12 predictors. Conversely, the expected size for BMA, where predictors impact equity returns only through the conditional volatility process is on average twice as high. For the latter, I observe a degree of affinity between average model size and the business cycle: Average model size increases on the onset of a recession and decreases following the recovery phase. However, it must be mentioned that this relationship is a bit more complex than has been stated. For instance, expected model size responds very aggressively to the Great Recession but less so to other recessions in the dataset. I also investigate the importance of individual predictors in the BMAs. Here, variables, such as the dividend yield and monthly S&P 500 realized volatility play a dominant role among the set of predictors. They consistently receive high inclusion probabilities, whereas earnings-to-price ratio and dividend-payout ratio become important only towards the end of the sample.

Lastly, I observe relative economic gains in favor of certain BMA models. In particular, following a similar approach to Campbell and Thompson (2008), Dangl and Halling (2012), Rapach and Zhou (2013), I find that an investor relying on predictions from a BMA, where predictors impact aggregate equity returns through the conditional mean as well as the conditional volatility process, gives utility gains between 2% and 4% per year over different out-of-sample periods relative to an investor that relies on predictions produced under the stochastic volatility benchmark. In comparison, an investor relying on predictions produced under BMAs, where only one of the predictive channels is turned on realizes only minor relative utility gains. I find that this result is due to the fact that an investor who relies on predictions from the first specification is better able to time the market by withdrawing at the onset of a recession and then moving back in to the market towards the beginning of a recovery.

The remainder of this study is organized as follows: Section 2 presents the econometric framework. Section 3 provides details regarding the study design. Here, I discuss the dataset and choice of the prior hyperparameter values. Out-of-sample results are presented and discussed in

[†] There are also a number of studies that use variables, such as changes in the price of crude oil, technical indicators and policy uncertainty measures as predictors, see for example, Driesprong *et al.* (2008), Neely *et al.* (2014), Brogaard and Detzel (2015). The suggested BMA framework can be applied including these data as well. However, the mentioned variables tend to cover a much shorter time-period. For example, Alquist *et al.* (2013) advise against using crude oil price data before 1972m12 because the price of crude oil varies little in this period and tends to exhibit a pattern resembling a step-function. Hence, if I were to incorporate crude oil price changes as a predictor, then the data would start from 1973m1. Likewise, the sample of Neely *et al.* (2014) start in 1951m12 due to data availability for the technical indicators. On the other hand, I choose to rely on the (Goyal and Welch 2008) dataset and the selected time-span for the following reasons: It makes results in this study qualitatively comparable with the mainstream literature that to a large degree also relies on this dataset, see for example, Dangl and Halling (2012), Rapach and Zhou (2013), Pan *et al.* (2020). Likewise, the data goes back to 1926m12, and by affording such a long historical time span, it contains important episodes, such as geopolitical tensions, crude oil and financial crises. This in turn, provides a very good opportunity to evaluate the predictive power afforded by one specification relative to another.

Section 4, whereas conclusions and future research directions are discussed in Section 5. Appendix A.1 provides details regarding implementation of the suggested dual Kalman filter. Finally, Appendix A.2 contains several sensitivity checks.

2. Econometric framework

I start by explaining how a given predictive model in the universe of models is specified and estimated. In Section 2.2, I discuss how density (point) predictions produced under different univariate models are combined to form the BMA density (point) prediction. Essentially, the suggested BMA averages over all possible combinations of the predictors that enter the model either through the conditional mean process, conditional volatility process or both. The weight attached to each combination among the total set of combinations is constructed using the individual model predictive likelihoods, which is a key Bayesian quantity for model comparison.[†] In fact, any quantity of interest is the weighted average of model-specific quantities with the model probabilities as weights.

A given time-varying predictive model in the universe of models can be expressed as:

$$y_t = \alpha_t + X_{t-1}\beta_t + \varepsilon_t, \quad \varepsilon_t \sim N \left(0, \underbrace{\exp(X_{t-1}\delta_t + h_t)}_{\sigma_t^2} \right), \quad (2.1)$$

$$\theta_t = \theta_{t-1} + \omega_t, \quad \omega_t \sim N(0, W_t), \quad (2.2)$$

$$\varphi_t = \varphi_{t-1} + \xi_t, \quad \xi_t \sim N(0, V_t), \quad (2.3)$$

where $\theta_t = (\alpha_t \parallel \beta_t)$ and $\varphi_t = (\delta_t \parallel h_t)$. In (2.2) and (2.3), the off-diagonal elements of W_t and V_t are non-zero. Likewise, y_t is the dependent variable at time t , $X_{t-1} = [X_{1,t-1}, \dots, X_{p,t-1}]$ is $1 \times p$ vector of predictors, ε_t is the time t innovation of the measurement equation, (2.1), that follows a Gaussian process with mean zero and conditional variance, $\sigma_t^2 = \exp(X_{t-1}\delta_t + h_t)$. Evidently, the interesting feature of $\exp(X_{t-1}\delta_t + h_t)$ is that it is a function of the unobserved log-volatility process, h_t , as well as the predictors, X_{t-1} , that impact the conditional distribution of y_t through the vector of time-varying coefficients, δ_t . Following seminal studies, such as Dangl and Halling (2012), the $(p+1) \times 1$ state equation vector, θ_t , consists of unobservable, time-varying regression coefficients and evolves according to (2.2). The same holds true for φ_t , which is also $(p+1) \times 1$. In (2.1), conditional innovation volatility follows the well-known stochastic volatility (SV) process. Moreover, in a similar fashion as Chan *et al.* (2013), Pettenuzzo *et al.* (2014), among

others, the conditional log-volatility at time t , h_t , evolves as a random-walk.[‡] The literature generally agrees that SV is more attractive than the widely used GARCH specification. For example, as noted in Johannes *et al.* (2014) and Koopman *et al.* (2016), compared to GARCH, the SV specification has an important advantage with regards to generating density predictions, namely, it produces fat-tailed marginal return distributions even though the innovation term follows a Gaussian distribution.[§] With regards to model estimation, the fact that one can rely on the linear state-space model simultaneously with a discount factor to filter out the latent log-volatilities also favors the SV specification. I provide more details regarding this point in Section 2.1.

Before proceeding further, I must make the reader aware of the following: As illustrated in (2.1), when $\beta_t \neq 0$ and $\delta_t \neq 0$, the same predictors enter into the conditional mean and volatility equations. Evidently, one could modify the model, such that different predictors are multiplied with β_t and δ_t . This is very easy in the context of a single model. However, with regards to model averaging, this approach becomes very cumbersome. This is because for a given set of predictors that enter the conditional mean equation, one would have to average over 2^p predictor combinations that enter the conditional volatility equation and vice versa. This in turn, increases the computation time drastically and more importantly does not allow one to assess the impact of isolating each channel. On the other hand, averaging over various models of the form (2.1) to (2.3), where different predictive channels are turned on or off by setting $\beta_{1:T} \neq 0$ ($\beta_{1:T} = 0$) or $\delta_{1:T} \neq 0$ ($\delta_{1:T} = 0$) makes model estimation easier while also allowing one to evaluate (at each out-of-sample observation), which extensions help improve prediction accuracy the most.

Accordingly, (2.2) and (2.3) imply that $\Upsilon_t = (\theta_t, \parallel \varphi)$ can drift to arbitrary high or low values, which is somewhat unattractive from a theoretical point of view. However, as argued in Dangl and Halling (2012), performing estimation using data at the monthly sampling frequency tends to mitigate any major concerns regarding the random-walk assumption.[¶] Furthermore, in a similar fashion as Dangl and

[†] As mentioned in studies, such as Geweke (2005), Geweke and Whiteman (2006), the predictive likelihood contains the out-of-sample prediction record of a model, making it an important criterion for model evaluation. This quantity automatically includes Occam's razor effect in that it penalizes highly parameter models that do not deliver improved predictive content. It is widely used to combine forecasts produced under different models, see for example, Kascha and Ravazzolo (2010) and Jore *et al.* (2010), among others.

[‡] I use the random-walk SV specification because it is the most general way of accounting for time-varying conditional volatility. Specifying an AR(1) structure on h_t very often signifies that I estimate the AR(1) coefficient very close to one. I can also refer the reader to Pettenuzzo *et al.* (2014) and a large number of studies in the macroeconomic literature, where the random-walk specification is relied on.

[§] As an example, I can refer the reader to Johannes *et al.* (2014), the marginal and predictive distributions of returns that integrate out the unobserved log-volatilities are a scale mixture of Normals, which is leptokurtotic.

[¶] As argued by the authors, such concerns might be more justified when updating at a lower sampling frequency, for example, quarterly. Furthermore, despite the raised concerns, the random-walk assumption is widely used in other fields of economics, such as macroeconomic forecasting and exchange rate prediction, see Meese and Rogoff (1983), Cogley and Sargent (2005), Primiceri (2005), Chan *et al.* (2013), among others. For instance, it is well-known that the random-walk model outperforms more sophisticated models with regards to predicting exchange rates. Likewise, Chan *et al.* (2013) introduce a model, where trend inflation is bounded. However, compared to the random-walk specification, the authors do not find substantial out-of-sample accuracy gains in favor of the bounded model.

Halling (2012), Byrne *et al.* (2018) at each time-period, I specify the degree of variation in θ_t and φ_t by evaluating, which values of W_t and V_t are most compatible with the data at hand via discount factors, see Section 2.1. By experimenting with several sets of discount factors, I observe that low discount factor values that in turn, imply values of W_t and V_t associated with the non-stationary regions of θ_t and φ_t do not receive much weight in the averaging process. Conversely, high values close to one do, meaning that one avoids unreasonable/explosive behavior of θ_t and φ_t . I also follow Dangl and Halling (2012) and experiment with imposing an autoregressive structure on Υ . However, results available upon request document that specifying an autoregressive (AR) structure on (2.2) and (2.3) does not lead to prediction accuracy gains relative to the random-walk specification implemented in this study.

In (2.1)–(2.3), when $\delta_t = 0$ in (2.1) for $t = 1, \dots, T$, one recovers the dynamic linear model suggested in West and Harrison (1997) and used in Dangl and Halling (2012).[†] Conversely, when $\beta_t = 0$ and $\delta_t \neq 0$ for $t = 1, \dots, T$, one has a model similar to Maheu *et al.* (2020), where X_{t-1} drives variations in y_t exclusively through its impact on the conditional volatility process. It is also possible that $\beta_t \neq 0$ and $\delta_t \neq 0$ for $t = 1, \dots, T$, meaning that predictors impact the dependent variable through the conditional mean as well as the conditional volatility process. Lastly, when the conditional covariance matrices, W_t and V_t , are fixed at zero for $t = 1, \dots, T$, one ends up with time-invariant predictive regressions. Here, one can again choose to set β and φ equal to zero or not.

2.1. Inference

In this section, I start by describing how a given model in the universe of models is estimated. Thereafter, in Section 2.2, I discuss how predictions produced under individual models are combined to form the BMA-based prediction. To estimate (2.1)–(2.3), I follow studies, such as Dangl and Halling (2012), Koop and Korobilis (2012) and proceed as follows: To begin with, one needs to give some sort of structure to W_t and V_t . Following the mentioned studies, I specify these matrices using very sensible approximations. Particularly, I assume that W_t (V_t) is proportional to the covariance of θ_t (φ_t), namely, Σ_t (Γ_t). More precisely, I assume that $W_t = (\lambda^{-1} - 1)\Sigma_{t-1}$ ($V_t = (\kappa^{-1} - 1)\Gamma_{t-1}$), where $0 < \lambda \leq 1$ ($0 < \kappa \leq 1$) is the well-known discount factor, see West and Harrison (1997) for a good introduction.[‡] Evidently, $\lambda = 1$ ($\kappa = 1$) corresponds to $W_t = 0$ ($V_t = 0$). Therefore, θ_t (φ_t) does not change between $t - 1$ and t . Conversely, decreasing λ (κ) below one implies that W_t (V_t) increases and θ_t (φ_t) is exposed to larger shocks, allowing the model to adapt more quickly to new information. Another way to interpret λ (κ) is that observations j periods in the past have weight λ^j (κ^j) in

determining the trajectory of θ_t (φ_t). Therefore, it is not surprising that the literature often refers to this as ‘aged-weighted estimation’.[§] For instance, $\lambda = 0.98$ means that W_t increases by 50% within 20 months, which corresponds to gradual time-variation in θ_t . On the other hand, fixing λ at 0.96 translates into 50% increase in 10 months, suggesting relatively more frequent changes in θ_t .[¶] Evidently, as $\lambda \rightarrow 0$, one ends up with a case, where only the most recent observations are of use for determining W_t . These dynamics also hold for φ_t . The choice of λ (κ) plays an important role with regards to model performance. A low λ (κ) implies that the model adapts more easily to data changes. However, this flexibility comes at great risk, namely, the model can become too flexible and as a result overfit the data.

Accordingly, specifying W_t and V_t for a given model in the universe of models with the help of discount factors implies a highly parsimonious structure, enabling one to avoid running Markov chain Monte Carlo (MCMC) simulations. Specifically, conditional on λ and κ , estimation of (2.1)–(2.3) is very fast, taking only a few seconds. Moreover, there is no need to re-estimate the model as new observations arrive because W_t and V_t for $t = 1, \dots, T$ are functions of λ and κ , respectively. On the other hand, readers familiar with MCMC techniques know quite well that estimation of (2.1)–(2.3) (or any other time-varying parameter model) using MCMC would be much more time consuming, especially in the context of recursive forecasting, and if the procedure has to be implemented on each model in the set. This is because at each iteration of the MCMC sampler, one would need to run the simulation smoother to generate the latent states from their conditional posteriors. This requires running a forward and a backward loop of size T each, see Carter and Kohn (1994), Kim and Nelson (1999), Durbin and Koopman (2012) for illustrations. Similar challenges exist if one were to model conditional volatility using GARCH dynamics.

Besides accounting for time-variation in the model parameters, one must also deal with $X_{t-1}\delta_t$ in (2.1) and the ‘extra’ state equation, φ_t . Here, I choose to adopt a similar strategy as Koop and Korobilis (2014) and use a dual linear Kalman filter, see Nelson and Stear (1976), Doz *et al.* (2011)

[§] Another way of interpreting λ (κ) is that it implies an effective window size of $1/(1 - \lambda)$ ($1/(1 - \kappa)$), which makes it comparable to windowing as suggested in Jazwinsky (1970).

[¶] Note, I assume that all elements in θ_t and φ_t lose information at the rate specified by λ and κ , respectively. Evidently, I could specify a separate discount factor for each element in θ_t and φ_t . This modification can be beneficial if the individual elements in θ_t and φ_t are losing information at drastically different rates. However, given the enormous number of computations required to accommodate this extension, I choose to maintain the current framework. Furthermore, as a robustness check, I compare filtered estimates of the conditional log-volatility process for BMA where $\beta_t \neq 0$ and $\delta_t = 0$ with BMA where $\beta_t \neq 0$ and $\delta_t \neq 0$. For the latter, h_t and δ_t lose information at the same rate. Hence, any eventual weaknesses (identified by different estimates of $h_{1:T}$) from the suggested approach would be very evident. Here, I observe that the filtered quantities are very similar, meaning that I do not face the problem that individual elements in φ_t are losing information at such different rates that would require me to change the approach. I must also mention that studies, such as Koop and Korobilis (2013) that estimate large time-varying parameter VAR models also use a single discount rate for all processes.

[†] The only difference is that in (2.1)–(2.3), when $\delta_t = 0$ for $t = 1, \dots, T$, conditional volatility innovation follows a stochastic volatility process, whereas it is homoscedastic in West and Harrison (1997), Dangl and Halling (2012).

[‡] The use of discount factors in the state-space literature goes back to Fagin (1964), Jazwinsky (1970).

for details.[†] Essentially, the idea is very simple: Given the value of $\tau = (\lambda, \kappa)'$, I first update θ_t given φ_t , and subsequently update φ_t given θ_t . Here, I use that (2.1)–(2.2) and (2.1)–(2.3) can individually be written in linear state-space form, and consequently φ_t and θ_t can be filtered out using the Kalman filter, respectively, see Appendix A.1 for details. The important point here is that $\tilde{e}_t = y_t - H_t \theta_{t|t}$, where $H_t = [1, X_{t-1}]$, $\theta_{t|t-1} = (\alpha_{t|t-1} \parallel \beta_{t|t-1})$ and $\alpha_{t|t-1} = E[\alpha_t | y_{1:t-1}]$ is used when updating φ_t , see (A1) in Appendix. This approach works because I update θ_t using the forgetting factor approach (which is an approximation-based method) and not directly estimating the model using simulation-based techniques, where one sequentially generates draws from the conditional posteriors of the latent states and the model parameters. This in turn, means that the Kalman filter provides a consistent estimate of $\theta_{t|t}$ even if there is a substantial amount of time-variation in θ_t . Moreover, simulation results using realistic data generating processes demonstrate that the suggested framework is able to recover the true latent processes. As an additional check, I consider (2.1)–(2.3), and estimate it using MCMC as well as the suggested approach. By comparing the estimates, I observe that the estimated latent processes follow each other very closely.[‡]

For the whole filtering process to work optimally, one must determine the value of $\tau = (\lambda, \kappa)'$ that is most compatible with the data at each time-period. The approach in this study is to follow Dangel and Halling (2012), Byrne *et al.* (2018) and besides averaging over the total number of predictor combinations also average over a range of values of, namely, $\tau \in \{\tau_1, \dots, \tau_d\}$. Particularly, for each $\tau_j \in \tau$, where it is allowed that $\kappa_j \neq \lambda_j$, I estimate 2^p predictive models that result from all possible combinations of the predictor vector, X_{t-1} .[§] Then, I use BMA to assign weights across individual models that differ in the selected variables and the vector of discount factors.

2.2. Bayesian model averaging

Let $\mathcal{M}_i$ denote a given combination of predictors from the set of $k = 2^p$ possible combinations available. Likewise, let $\tau_j = (\lambda_j, \kappa_j)'$, where it is possible that $\lambda_j \neq \kappa_j$ denote a given set of discount factors from the grid, $\{\tau_1, \dots, \tau_d\}$. This implies a total of $m = d \cdot k$ combinations at each time-period. Henceforth, let $p(\mathcal{M}_i | \tau_j, y_{1:t})$ denote the probability of model $\mathcal{M}_i$ conditional on $\tau_j \in \{\tau_1, \dots, \tau_d\}$ and information up to time t :

$$p(\mathcal{M}_i | \tau_j, y_{1:t}) = \frac{f(y_t | \mathcal{M}_i, \tau_j, y_{1:t-1}) p(\mathcal{M}_i | \tau_j, y_{1:t-1})}{\sum_{i=1}^k f(y_t | \mathcal{M}_i, \tau_j, y_{1:t-1}) p(\mathcal{M}_i | \tau_j, y_{1:t-1})}, \quad (2.4)$$

[†] As an alternative to this approach, I could use the Extended Kalman filter. However, I find this approach to be relatively unstable. I must also mention that Koop and Korobilis (2014) reach a similar conclusion regarding the Extended Kalman filter in their application.

[‡] With the term ‘realistic’, I mean that I simulate data, where the unobserved processes vary in a similar fashion as when the model is estimated on the Goyal and Welch (2008) data.

[§] Note, given the definition in (2.1), an intercept is always included in the conditional mean equation.

where $f(y_t | \mathcal{M}_i, \tau_j, y_{1:t-1})$ is the predictive density of y_t conditional on $\mathcal{M}_i$ and τ_j . Using (2.4), one can easily integrate out the effects of $\mathcal{M}_i$ and τ_j to obtain the BMA-based prediction. For example, by summing over the combinations of predictors and degrees of instability, the BMA predictive density is given as:

$$f(y_{t+1} | y_{1:t}) = \sum_{j=1}^d f(y_{t+1} | \tau_j, y_{1:t}) p(\tau_j | y_{1:t}), \quad (2.5)$$

where $p(\tau_j | y_{1:t})$ in (2.5) is computed as:

$$p(\tau_j | y_{1:t}) = \frac{f(y_t | \tau_j, y_{1:t-1}) p(\tau_j | y_{1:t-1})}{\sum_{j=1}^d f(y_t | \tau_j, y_{1:t-1}) p(\tau_j | y_{1:t-1})} \quad (2.6)$$

and

$$f(y_{t+1} | \tau_j, y_{1:t}) = \sum_{i=1}^k f(y_{t+1} | \mathcal{M}_i, \tau_j, y_{1:t}) p(\mathcal{M}_i | \tau_j, y_{1:t}) \quad (2.7)$$

where $f(y_{t+1} | \mathcal{M}_i, \tau_j, y_{1:t})$ in (2.7) is the predictive density of y_{t+1} generated conditional on τ_j , $\mathcal{M}_i$ and $y_{1:t}$. The BMA-based predictive mean of y_{t+1} conditional on $y_{1:t}$ can be computed in a same manner. Instead of averaging over all possible combinations of predictors, one could also select the specification with the highest posterior model probability and use predictions associated with it to predict. However, I generally find that model averaging and model selection lead to similar conclusions. Therefore, to be as concise as possible and save space, I focus only on averaging in this study. However, results based on Bayesian model selection are available upon request.

The output from the BMA can also be used to compute several other interesting quantities, which prove to be very useful in the empirical analysis. For example, I use (2.4), (2.6) and let

$$p(\mathcal{M}_i | y_{1:t}) = \sum_{j=1}^d p(\mathcal{M}_i | \tau_j, y_{1:t}) p(\tau_j | y_{1:t}), \quad (2.8)$$

denote the probability of model $\mathcal{M}_i$ conditional on $y_{1:t}$ after the effects of parameter instability have been integrated out. In a similar fashion as Dangel and Halling (2012), one can use (2.8) to compute $\sum_{i=1}^k 1_{(i \in l)} p(\mathcal{M}_i | y_{1:t})$, $l = 1, \dots, p$, where $1_{(i \in l)}$ is an indicator function taking the value of either 0 or 1 and l denotes the l th predictor. This measure, which I henceforth refer to as the inclusion probability of predictor l is a very useful tool to (ex post) evaluate how much support the predictor of interest receives at each time-period.

Finally, let $Size^{(i)}$ denote the number of predictors in $\mathcal{M}_i$, excluding the constant. The average number of predictors in the BMA at each time-period can easily be computed as:

$$E[Size | y_{1:t}] = \sum_{i=1}^k Size^{(i)} p(\mathcal{M}_i | y_{1:t}). \quad (2.9)$$

This quantity displays the degree of shrinkage over time.

3. Study design

In this section, I outline how the suggested BMA framework is operationalized on the data. BMA over (2.1)–(2.3), where $\beta_{1:T} \neq 0$, $\delta_{1:T} = 0$ is similar to the BMA suggested in Dangl and Halling (2012).[†] Henceforth, I refer to this as BMA incl. TVP $\beta_{1:T} \neq 0$, $\delta_{1:T} = 0$, indicating that I consider BMA with time-varying (TVP) model parameters, where the predictive impact occurs exclusively through the conditional mean process. Accordingly, I can easily have a BMA incl. TVP, where $\beta_{1:T} = 0$ and $\delta_{1:T} \neq 0$ or where $\beta_{1:T} \neq 0$ and $\delta_{1:T} \neq 0$. This in turn, enables me to evaluate the predictive power afforded by changing the predictive channel or have the unrestricted model, where all predictive channels are turned on. Finally, I also consider BMAs, where $\lambda_{1:T} = \kappa_{1:T} = 1$, which I denote as BMA excl. TVP. Here, I consider cases where $(\beta \neq 0, \delta = 0)$, $(\beta = 0, \delta \neq 0)$ and $\beta \neq 0, \delta \neq 0$.

As previously mentioned, in a recent study, Pan *et al.* (2020) suggest a simple method to impose economic constraints on univariate predictive regressions of aggregate equity returns. Interestingly, unlike studies, such as Campbell and Thompson (2008), Pettenuzzo *et al.* (2014), the constraint is implemented directly on the predictor and not on the regression coefficient or the sign of the prediction. Particularly, they introduce a simple truncation of the predictor of interest, $X_{l,t}$, and define the following non-linear transformation of $X_{l,t}$ as:

$$X_{l,t} = \begin{cases} X_{l,t} & \text{if } X_{l,t} > \max(X_{l,t-1}, \dots, X_{l,t-24}) \\ & \text{or } X_{l,t} < \min(X_{l,t-1}, \dots, X_{l,t-24}) \\ 0 & \text{otherwise.} \end{cases} \quad (3.1)$$

In other words, the predictor is set to zero whenever its value falls within its past 24-month high and low.[‡] The authors demonstrate that a simple model average using (3.1) results in point accuracy gains at the population level relative to the constrained methods suggested in Campbell and Thompson (2008), Pettenuzzo *et al.* (2014). Implementing this extension is very easy in the suggested framework. Henceforth, I refer to this specification as BMAR, which stands for BMA with restrictions on the predictors via (3.1).

3.1. Prior and parameter choice

Let subscript (m) signify the quantity of interest associated with model m among the $d \cdot k$ models. The recursions presented in Section 2.2 require specifying priors on $\theta_0^{(m)}$ and $\varphi_0^{(m)}$ for $m = 1, \dots, d \cdot k$. I also have to choose $p(\mathcal{M}_i | \delta_j, y_0)$ for $i = 1, \dots, k$ and $p(\tau_j | y_0)$ for $j = 1, \dots, d$. The former is used to compute (2.4), whereas the latter is used to compute (2.6). With regards to the priors on $\theta_0^{(m)}$ and $\varphi_0^{(m)}$, I specify non-informative priors. Specifically, I assume that $p(\theta_0^{(m)} | y_0)$ and $p(\varphi_0^{(m)} | y_0)$ follow a Gaussian distribution centered at zero with variance equal to $\Sigma_0^{(m)} = 10I$

and $\Gamma_0^{(m)} = 10I$, respectively. Likewise, following Dangl and Halling (2012), Byrne *et al.* (2018), I let $p(\mathcal{M}_i | \tau_j, y_0) = 1/k$, $i = 1, \dots, k$, and $p(\tau_j | y_0) = 1/d$, $j = 1, \dots, d$, such that initially all model combinations receive equal probability. I must also choose a set of values for τ_j . Again, following Dangl and Halling (2012), I consider several combinations of $\lambda \in [0.96, 0.97, \dots, 1]$ and $\kappa \in [0.96, 0.97, \dots, 1]$. More precisely, for each individual value in $\lambda \in [0.96, 0.97, \dots, 1]$, I consider the grid $\kappa \in [0.96, 0.97, \dots, 1]$, leading to a total 25 combinations to consider.[§]

Overall, I find that results are robust to the choice of the prior hyperparameter values, initial model probabilities and the considered values of the discount factors. However, this should not be very surprising. To begin with, I rely on a long time-series of historical data. The data itself starts at 1926m12, and the earliest out-of-sample period starts at 1947m1 (20 years later), meaning that there is plenty of time for the model to adapt to different patterns in the data before the out-of-sample evaluation starts. Consequently, any eventual effects of the prior hyperparameter values die out before I start evaluating the out-of-sample performance of the models.[¶] Likewise, decreasing the lower bounds for the specified values of λ and κ does not change anything as results in Section 4.3 indicate that at any given point in time, the BMA estimate of λ (κ) never falls below 0.97 (0.98). Nevertheless, in Appendix A.2, I provide examples, demonstrating robustness of the results to the choice of these input parameters.

3.2. Data

In a similar fashion as Dangl and Halling (2012), Rapach and Zhou (2013), Beckmann and Schüssler (2014), Pettenuzzo and Ravazzolo (2016), Pan *et al.* (2020) among many others, my analysis relies on monthly returns on the Standard & Poor's (S&P) 500 index along with a set of 12 well-known economic predictors using an updated version of Goyal and Welch (2008) data. The sample itself runs from 1926m12 through 2018m12.^{||}

Table 1 provides a list of the predictors used in the analysis, along with a short description on how they are constructed. I must mention that contrary to Goyal and Welch (2008), the variable, dividend-to-price ratio is excluded from this study since it is almost perfectly correlated with the dividend yield. The same holds for the term spread, which is the difference between lty and tbl. When constructing and applying these predictors, I strictly follow the above mentioned

[§] Although this might seem daunting from a computational viewpoint, I must mention that I can easily use parallel computing. For example, in Matlab, I can simply use `parfor`, which allows one to execute for-loops in parallel.

[¶] By conducting various experiments, I find that it takes at maximum 150 months for the effects of the prior hyperparameters to die out.

^{||} The data is extracted from Amit Goyal's website: <http://www.hec.unil.ch/agoyal/>. The choice of the sample-period is driven purely by concerns of data availability. For example, returns on the S&P 500 index are available from the late 1800s. However, a large number of the economic predictors are only available from the early 1900s.

[†] The difference is that I account for time-varying innovation volatility, whereas Dangl and Halling (2012) assume constant volatility.

[‡] It is worth mentioning that the econometric frameworks suggested in Campbell and Thompson (2008), Pettenuzzo *et al.* (2014) rely on models with a single predictor. Recently, Chib and Zeng (2016) extend the framework of Pettenuzzo *et al.* (2014) to a multivariate setting and find predictive gains of similar magnitude.

Table 1. Economic variables used as predictors.

Label	Description
Dy	dividend yield defined as the difference between the log of dividends on the S&P 500 index and the log of lagged S&P 500 index.
Ep	earnings-to-price ratio defined as the difference between the log of earnings and the log of the S&P 500 index.
Dpayr	dividend-payout ratio defined as the difference between the log of dividends and the log of earnings.
Bmr	book-to-market ratio defined as the ratio of book value at the end of the previous year divided by the market value.
Svar	S&P 500 monthly log-realized volatility.
Ntis	net equity expansion defined as the moving sums of net issues by NYSE listed stocks to the total market capitalization.
Tbl	3-month U.S. treasury bills rate.
Lty	long-term government bond yields.
Ltr	long-term return on government bonds.
Dfr	default return spread defined as the difference between returns on long-term corporate bonds and government bonds.
Dfy	default yield spread defined as the difference between BAA-rated and AAA-rated corporate bond yields.
Inf	CPI inflation rate.

Note: This table presents the financial and economic variables used in the empirical analysis. They are downloaded from Amit Goyal's website: <http://www.hec.unil.ch/agoyal/>.

studies. I do this despite evidence of persistence in certain variables.[†]

3.3. Prediction evaluation methodology

I follow the previously mentioned studies and evaluate the accuracy of point predictions produced under the different models using the mean square error (MSE) criterion. However, a point prediction measures the central tendency of the target variable, or the best prediction, but it is silent about the uncertainty that is associated with producing it. A density prediction on the other hand takes into account the whole shape of the conditional distribution, thereby containing a complete description of future outcomes of the variable of interest. It also allows the researcher to put greater emphasis on a particular region of the conditional distribution, such as the center, right and/or the left tail. Therefore, I also evaluate density predictions from the range of models using

[†] Results from the Augmented (Dickey and Fuller 1981) test suggest that while some predictors, such as the dividend yield are very persistent, the persistence is not so severe so as to require alternative inference frameworks, such as Tourus *et al.* (2004), Campbell and Yogo (2006), Jansson and Moreira (2006). I state this because there is evidence against the unit root null hypothesis for all variables. Furthermore, given that I work with a dataset containing over 1000 observations eases concerns regarding the impact of predictor endogeneity. This is because the bias caused by neglecting potential predictor endogeneity vanishes with sample size, see for example, Stambaugh (1999), Westerlund and Narayan (2012). In fact, Monte Carlo results performed in the latter study support our argument.

Table 2. Weight functions for the wCRPS criterion.

Emphasis	uniform	center	tails	right tail	left tail
$w(z)$	1	$\phi(z)$	$1 - \phi(z)/\phi(0)$	$\Phi(z)$	$1 - \Phi(z)$

Note: This table reports various weight functions, which I use to compute the weighted continuous ranked probability score (wCRPS) criterion. $\phi(z)$ and $\Phi(z)$, denote the probability density and cumulative distribution functions of a standard Gaussian distribution.

the weighted continuous ranked probability score (wCRPS) criterion suggested in Gneiting and Ranjan (2011).[‡]

Essentially, wCRPS measures the average absolute distance between the empirical cumulative distribution function (CDF) of y_{t+1} , which is simply a step function in y_{t+1} , and the predicted CDF that is associated with the model's predictive density. Gneiting and Ranjan (2011) demonstrate that comparison between the empirical and predicted CDF can be weighed by a function that emphasizes a certain area of the underlying predictive density, for example, the center, tails, right or the left tail separately. The wCRPS criterion is defined as follows:

$$\text{wCRPS}_{t+1} = \int_{-\infty}^{\infty} w(z) (F(z) - I(y_{t+1} \leq z))^2 dz, \quad (3.2)$$

where $w(z)$ is the weight function, $F(z)$ is the cumulative distribution function of the model of interest evaluated at z , and $I(\cdot)$ takes the value 1 if $y_{t+1} \leq z$, otherwise it is equal to 0. With regards to $w(z)$, the simplest case is obviously $w(z) = 1$, meaning that one attaches the same weight to each region of the conditional distribution. Besides $w(z) = 1$, I also consider the weights reported in table 2 to evaluate how density predictions produced under the considered models fare when the center, tails, right and left tail are emphasized, respectively.

In order to quantify the magnitude of relative density prediction accuracy gains afforded by the predictive model of interest, I report the average (3.2) produced under the model of interest relative to the benchmark. If the corresponding ratio, namely, avCRPS is less than one, then I conclude that density predictions produced under the predictive model of interest are more accurate than density predictions produced under the benchmark. Furthermore, to determine whether density (point) predictions produced under the predictive model of interest are statistically significant more accurate than density (point) predictions produced under the benchmark or vice versa, I use the Diebold and Mariano (1995) test of equal finite-sample predictive ability. This test is given as:

$$DM = \frac{\Delta \bar{L}}{\hat{\sigma}_{HAC} / \sqrt{n}}. \quad (3.3)$$

In (3.3) $\Delta \bar{L} = n^{-1} \sum_{i=1}^n \Delta L_i$, where ΔL_i denotes the wCRPS or the squared point prediction error difference between the benchmark and the model of interest at out-of-sample observation i , and n is the number out-of-sample observations. Lastly, $\hat{\sigma}_{HAC}^2$ is a Heteroskedasticity and Autocorrelation Consistent (HAC) estimator of the asymptotic variance,

[‡] I refer the reader to their study for details regarding the advantages of wCRPS relatively to the widely used log-score criterion.

$\sigma^2 = \text{var}[\sqrt{n}\Delta\bar{L}]$. The null hypothesis of the Diebold and Mariano (1995) test is equal unconditional predictive ability, namely, $E[\Delta L_i] = 0$ versus the alternative, $E[\Delta L_i] \neq 0$.[†]

Although the main focus of this study is on density prediction comparison between different models, I also evaluate the economic power afforded by them. I follow studies, such as Campbell and Thompson (2008), Dangl and Halling (2012), Johannes *et al.* (2014) and consider an investor with mean-variance preference. The investor's utility function at time t is given as $U_t = E[r_{p,t}] - (\gamma/2)\text{var}(r_{p,t})$, where $r_{p,t}$ is the time t return on the portfolio. The portfolio itself is composed of the return on the S&P 500 index and the risk-free rate (measured using the 3-month U.S. t-bill rate), r_t^f . The investor attaches the weight, ϖ_t , to the risky asset and $(1 - \varpi_t)$ to the risk-free rate, r_t^f , implying that the portfolio return is $\varpi_t r_{p,t} + (1 - \varpi_t)r_t^f$. In a similar fashion as Campbell and Thompson (2008), Dangl and Halling (2012), the relative degree of risk aversion, γ , is fixed at three and six, respectively.[‡] Given information at time t , the ex ante optimal weight attached to the risky asset at time $t + 1$ is given as:

$$\varpi_{t+1}^* = \frac{1}{\gamma} \left(\frac{\hat{y}_{t+1} - r_t^f}{\hat{\sigma}_{t+1}^2} \right). \quad (3.4)$$

As evidenced from (3.4), for certain BMAs, the predictors introduced in Section 3.2 impact ϖ_{t+1}^* , through the conditional mean channel (i.e. $\hat{y}_{t+1}$) as well as the conditional volatility channel (i.e. $\hat{\sigma}_{t+1}^2$). I also follow Dangl and Halling (2012) and limit the share invested into the risky asset to be between 0 and 1.5 in order to preclude short sales and to prevent more than 50% leverage. If $\bar{U}^l = n^{-1} \sum_{j=1}^n U_j^l$ is greater than $\bar{U}^b = n^{-1} \sum_{j=1}^n U_j^b$, I conclude that the former specification is preferred from an economic viewpoint and vice versa.

3.4. Discussion

This section provides a discussion regarding why allowing the predictors to impact aggregate equity returns through the conditional volatility channel could help further improve relative density prediction accuracy. As previously mentioned, a density prediction takes into account the whole shape of the conditional return distribution. Therefore, incorporating information from variables that help drive variations in higher moments of aggregate equity returns can indeed be useful with respect to further improving density prediction accuracy. Interestingly, a number of recent studies find that certain economic variables that help predict aggregate equity returns out-of-sample also help forecast aggregate equity return volatility out-of-sample, see for example, Christiansen *et al.* (2012), Paye (2012), Nonejad (2017).[§] These variables include valuation ratios, such as the dividend yield and

earnings-to-price ratio as well as bond yields, rates of interest and measures of corporate payout.[¶] The mentioned studies also find evidence in favor of combining forecasts produced under different models, for example through BMA. In fact, using a similar BMA as Dangl and Halling (2012), Nonejad (2017) finds that accounting for parameter instability and model uncertainty helps improve out-of-sample point and density forecasts accuracy of monthly S&P 500 realized volatility compared to other alternative models. Lastly, Paye (2012), Nonejad (2017), among others establish a link between predictive ability and the onset of recessions. Given these findings, it is highly likely that besides the conditional mean channel, one could obtain further density prediction accuracy gains by also allowing the predictors to impact aggregate equity returns through the conditional volatility channel.

To elaborate on this point, I provide a simple example. I consider (2.1)–(2.3), where $X_{t-1} = X_{1,t-1}$ with $X_{1,t-1}$ corresponding to the divided yield. I then estimate the model, where $\beta_{1:T} \neq 0$, $\delta_{1:T} = 0$ and $\beta_{1:T} \neq 0$, $\delta_{1:T} \neq 0$, respectively. In panels (a) and (b) of figure 1, I plot the estimate of $\sigma_{1:T}$ for (2.1) under the two models. In these panels, vertical lines indicate the business cycle i.e. the point at which an economic expansion transitions into a recession and subsequently recovers using National Bureau of Economic Research (NBER) business cycle dating. As displayed in the panels, both conditional innovation volatility estimates are at the same level. However, compared to panel (a), there is a more clear connection between spikes in conditional volatility and the onset of recessions for (2.1)–(2.3), where $\beta_{1:T} \neq 0$, $\delta_{1:T} \neq 0$. This is very evident in the context of the recessions early in the sample, the early/mid 1970s and Great Recession. In other instances, the connection is less evident.

What are the implications of results in panels (a) and (b) with respect to generating accurate density predictions? To answer this question, I consider the same models and generate the conditional distribution of y_t at 1974m6 given information up to 1974m5. As displayed in panels (a) and (b), during this time-period, a recession has shortly begun, and one observes differences between $\hat{\sigma}_t$ s produced under (2.1)–(2.3), where $\beta_{1:T} \neq 0$, $\delta_{1:T} = 0$ compared to $\hat{\sigma}_t$ s produced under (2.1)–(2.3), where $\beta_{1:T} \neq 0$, $\delta_{1:T} \neq 0$. In panels (c) and (d), I display the corresponding predictive densities. In these panels, the dark vertical lines indicate the realized value of y_t at 1974m6. Accordingly, the return on the S&P 500 index is negative. Moreover, by comparing the panels, it is apparent that results are more positive for (2.1)–(2.3), where $\beta_{1:T} \neq 0$, $\delta_{1:T} \neq 0$.

while aggregate equity returns are not. According to Paye (2012), this conjures a distinct sense of *déjà vu*.

[¶]Conceptually, the conditional variance of equity returns depends upon the conditional variance of future cash flows, the conditional variance of the discount rates and conditional covariance between these two components. This implies that channels through which the mention economic variables can generate variations in aggregate equity return volatility include (but are not limited to): Shocks to expected cash flows and the discount rates, see Bansal and Yaron (2004), Mele (2005, 2007), learning effects related to investor's uncertainty about fundamentals, see Veronesi (1999), Timmermann (1993, 1996), Brennan and Xia (2001), and amplification of shocks to asset markets via financial intermediation, see for example, Brunnermeier and Pedersen (2009).

[†]I have to make the reader aware that contrary to this study, the majority of studies on equity return predictability rely on the Clark and West (2007) test, which evaluates the evidence of population-level predictability and not finite-sample predictive ability.

[‡]As evidenced from (3.4), a higher value of γ implies that the risky asset is assigned a lower weight in the portfolio.

[§]Paye (2012) is essentially the first study to notice this phenomenon despite the fact that aggregate equity return volatility is persistent

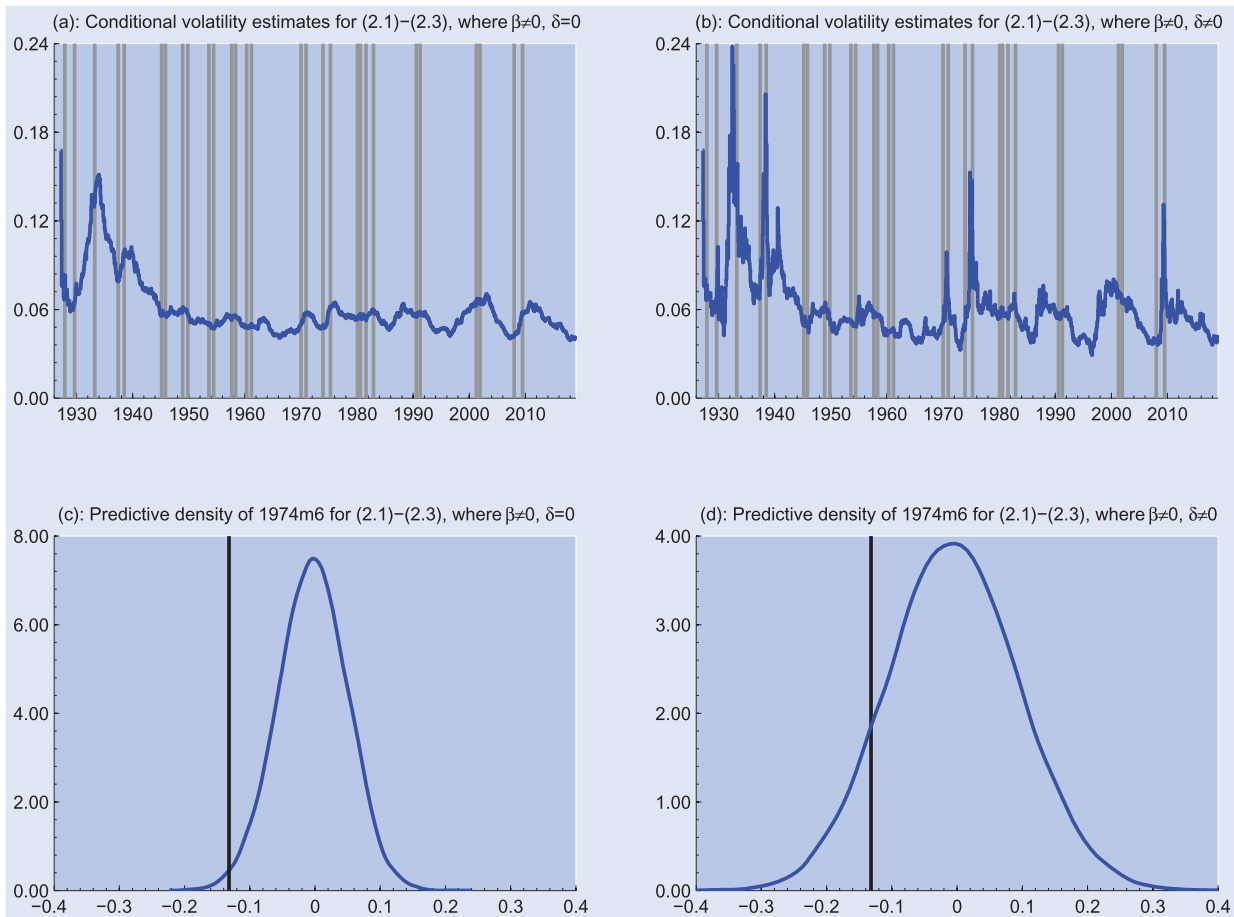


Figure 1. Conditional innovation volatility estimates and examples of predictive densities for different models.

In panels (a) and (b), I display conditional volatility estimates for (2.1)–(2.3), where $X_{t-1} = X_{1,t-1}$ and $X_{1,t-1}$ corresponds to the divided yield. Vertical lines indicate economic recessions using National Bureau of Economic Research (NBER) business cycle dating. In panels (c) and (d), I display the generated conditional return distributions for 1974m6 given information up to 1974m5 under the same models. The vertical lines indicate the realized value of the dependent variable at 1974m6.

Here, compared to panel (c), the conditional return distribution is flatter and more centered around the realized outcome. I also compute $wCRPS_{1974m6}$ for each model. Here, compared to (2.1)–(2.3), where $\beta_{1:T} \neq 0, \delta_{1:T} = 0$, the accuracy gains in favor of (2.1)–(2.3), where $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$ are between 2% to 4% depending on which area of the conditional distribution is emphasized.

4. Out-of-sample analysis

The starting point of the analysis is the different BMA incl. (excl.) TVP models. Counterparts to these BMAs, where predictors are transformed via (3.1) are labeled as BMAR. In a similar fashion as Pan *et al.* (2020), SMA denotes the simple mean average across all possible combinations of the predictors. I consider two versions of this model. To begin with, I assume that θ and h vary over time. Next, I assume that they are time-invariant.

I also consider alternative models based on well-known classical selection techniques. To start with, I consider model selection using the BIC criterion. Particularly, at each out-of-sample observation, I use point (density) prediction from the model combination (among the 2^p) with the smallest

BIC. Here, models are recursively estimated using a rolling window OLS approach, where the window length is fixed at five years. The final model is the ridge regression, see also Groen *et al.* (2013). Essentially, it is a simple linear regression of y_t on all of the predictors,

$$y_t = \tilde{\theta}_0 + X_{t-1}\tilde{\theta} + e_t, \quad e_t \sim N(0, \sigma_e^2),$$

where $\tilde{\theta} = (\sum_{t=2}^T X'_{t-1}X_{t-1} + \psi I_p)^{-1}(\sum_{t=2}^T X'_{t-1}y_t)$ and ψ is the scalar shrinkage factor. To achieve good performance, I follow De Mol *et al.* (2008), Groen *et al.* (2013) and set $\psi = 12$, such that the degree of shrinkage is proportional to the number of regressors.[†] As benchmark, I consider the following stochastic volatility model:

$$y_t = \mu_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, \exp(h_t)), \quad (4.1)$$

$$\mu_t = \mu_{t-1} + \eta_t, \quad \eta_t \sim N(0, \sigma_\mu^2), \quad (4.2)$$

$$h_t = h_{t-1} + v_t, \quad v_t \sim N(0, \sigma_h^2). \quad (4.3)$$

Note, this benchmark is different from the historical average benchmark (often estimated using the expanding window

[†] The ridge regression is also recursively estimated using a rolling window OLS approach. The window length is fixed at five years.

Table 3. Statistical evaluation results over the 1947m1–2018m12 out-of-sample period.

Specification	incl. TVP		excl. TVP		incl. TVP		excl. TVP	
Model	TU	DM	TU	DM	avCRPS	DM	avCRPS	DM
BMA $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.01	0.44	1.01	0.46	1.00	0.48	1.05	0.00
BMA $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	1.00	0.12	0.99	0.25	0.99	0.03	0.99	0.00
BMA $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.00	0.82	1.01	0.54	1.00	0.81	1.00	0.86
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.01	0.29	1.02	0.06	1.00	0.27	1.05	0.00
BMAR $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	1.00	0.26	0.99	0.24	1.01	0.21	1.01	0.02
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.00	0.54	1.02	0.08	1.01	0.10	1.03	0.00
SMA	1.01	0.77	1.01	0.36	1.00	0.82	1.06	0.00
BIC			1.04	0.02			1.07	0.00
Ridge			1.01	0.32			0.99	0.35
	avCRPS-center	DM	avCRPS-center	DM	avCRPS-tails	DM	avCRPS-tails	DM
BMA $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.00	0.47	1.01	0.11	1.00	0.50	1.06	0.00
BMA $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	0.99	0.03	1.00	0.79	1.00	0.90	0.98	0.00
BMA $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.00	0.78	1.00	0.66	1.00	0.73	1.00	0.91
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.00	0.51	1.01	0.04	1.00	0.21	1.06	0.00
BMAR $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	1.00	0.16	1.00	0.70	1.01	0.17	1.02	0.02
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.00	0.72	1.01	0.02	1.01	0.08	1.04	0.00
SMA	1.00	0.98	1.01	0.01	1.00	0.76	1.08	0.00
BIC			1.02	0.00			1.08	0.00
Ridge			1.01	0.28			0.99	0.13
	avCRPS-right tail	DM	avCRPS-right tail	DM	avCRPS-left tail	DM	avCRPS-left tail	DM
BMA $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.01	0.22	1.06	0.00	1.00	0.98	1.03	0.00
BMA $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	1.00	0.55	0.99	0.04	0.99	0.10	0.99	0.03
BMA $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.00	0.58	1.00	0.56	1.00	0.92	0.99	0.62
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.01	0.02	1.07	0.00	1.00	0.73	1.03	0.00
BMAR $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	1.01	0.31	1.01	0.04	1.01	0.27	1.01	0.22
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.01	0.16	1.04	0.00	1.01	0.23	1.02	0.02
SMA	1.01	0.46	1.08	0.00	1.00	0.83	1.05	0.00
BIC			1.07	0.00			1.06	0.00
Ridge			0.99	0.32			1.00	0.66

Note: In the above table, TU (Theil's U) is defined as the MSE produced under the model of interest relative to the MSE produced under the stochastic volatility benchmark. Likewise, avCRPS reports the average (3.2) with the specified weight reported in table 2 produced under the model of interest relative to the avCRPS with the specified weight produced under the stochastic volatility benchmark.

approach) widely used in a number of studies. I believe it is important to consider (4.1)–(4.3) as benchmark because doing so, I can better isolate the predictive power afforded by each BMA extension. Conversely, if I were to use the historical average benchmark, then I could not firmly conclude whether relative density prediction accuracy gains from the BMAs are due to parameter instability, heteroscedasticity, model uncertainty or the predictive channels. Moreover, as displayed in panels (a) and (b) of figure 1, there is evidence of time-varying conditional innovation volatility. Therefore, with regards to evaluating the degree of relative density prediction accuracy gains afforded by the predictive model of interest, it is important that the benchmark model accounts for the fact that the conditional return distribution will indeed have changing variance over time. Lastly, studies such as Campbell and Thompson (2008), Beckmann and Schüssler (2014) also rely on similar heteroscedastic benchmarks. The former does so in an ad-hoc fashion by estimating the historical average benchmark over rolling windows of observations (5 year windows). The latter directly specifies time-variation in the conditional innovation volatility process in a somewhat similar fashion as in this study.

The predictive power of the models relative to (4.1)–(4.3) is evaluated across several out-of-sample periods, namely, 1947m1–2018m12, 1965m1–2018m12, 1976m1–2018m12, 1988m1–2018m12, 2000m1–2018m12 and 2010m1–2018m12.

The choice of these out-of-samples is driven by discussions in Goyal and Welch (2008), Dangl and Halling (2012), Paye (2012), among others. It is worth mentioning that the authors provide rigorous justifications for relying on these out-of-sample periods.[†] I refer the reader to the mentioned studies for details.

4.1. Results

Tables 3–8 present prediction results across the mentioned out-of-sample periods. More precisely, for each model and out-of-sample period, the tables present Theil's U (TU) defined as the MSE produced under the model of interest relative to the benchmark, and the average wCRPS with the weight function, $w(z)$, specified in table 2 produced under the model of interest relative to the stochastic volatility benchmark, avCRPS. Finally, the columns labeled DM report p -values from the Diebold and Mariano (1995) test of equal finite-sample predictive ability.

[†] For instance, Dangl and Halling (2012) argue that a common result in the equity return predictability literature is that evidence of out-of-sample predictability is to some extent driven by a few exceptional return observations. Therefore, 1976m1–2018m12 and 1988m1–2018m12 start immediately after the crude oil price shocks of the early 1970s and the stock market crash in 1987.

Table 4. Statistical evaluation results over the 1965*m*1–2018*m*12 out-of-sample period.

Specification	incl. TVP		excl. TVP		incl. TVP		excl. TVP	
Model	TU	DM	TU	DM	avCRPS	DM	avCRPS	DM
BMA $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.01	0.55	1.00	0.77	1.00	0.48	1.03	0.00
BMA $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	1.00	0.12	0.99	0.18	0.99	0.04	0.99	0.00
BMA $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.00	0.89	1.00	0.97	1.00	0.88	0.99	0.33
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.01	0.40	1.02	0.12	1.00	0.41	1.04	0.00
BMAR $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	1.00	0.30	0.99	0.19	1.01	0.21	1.00	0.68
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.00	0.71	1.02	0.16	1.01	0.10	1.02	0.03
SMA	1.00	0.89	1.00	0.95	1.00	0.98	1.04	0.00
BIC			1.03	0.06			1.04	0.00
Ridge			1.02	0.30			1.00	0.79
	avCRPS-center	DM	avCRPS-center	DM	avCRPS-tails	DM	avCRPS-tails	DM
BMA $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.00	0.48	1.00	0.50	1.00	0.50	1.03	0.00
BMA $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	0.99	0.09	1.00	0.62	1.00	0.94	0.98	0.00
BMA $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.00	0.52	0.99	0.14	1.00	0.73	0.99	0.42
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.00	0.59	1.01	0.07	1.00	0.37	1.04	0.00
BMAR $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	1.00	0.11	1.00	0.91	1.01	0.17	1.00	0.68
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.00	0.88	1.01	0.04	1.01	0.07	1.02	0.04
SMA	1.00	0.85	1.01	0.09	1.00	0.92	1.04	0.00
BIC			1.02	0.03			1.04	0.00
Ridge			1.01	0.14			1.00	0.87
	avCRPS-right tail	DM	avCRPS-right tail	DM	avCRPS-left tail	DM	avCRPS-left tail	DM
BMA $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.01	0.24	1.05	0.00	1.00	0.92	1.01	0.14
BMA $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	1.01	0.41	0.99	0.26	0.99	0.10	0.98	0.05
BMA $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.00	0.67	1.00	0.68	1.00	0.90	0.98	0.26
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.01	0.04	1.05	0.00	1.00	0.75	1.02	0.01
BMAR $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	1.01	0.27	1.01	0.53	1.01	0.36	0.99	0.41
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.01	0.10	1.02	0.03	1.01	0.35	1.01	0.23
SMA	1.01	0.44	1.05	0.00	0.99	0.61	1.03	0.01
BIC			1.05	0.00			1.02	0.01
Ridge			1.00	0.92			1.00	0.74

Note: In the above table, TU (Theil's U) is defined as the MSE produced under the model of interest relative to the MSE produced under the stochastic volatility benchmark. Likewise, avCRPS reports the average (3.2) with the specified weight reported in table 2 produced under the model of interest relative to the avCRPS with the specified weight produced under the stochastic volatility benchmark.

I start by discussing results over the 1947*m*1–2018*m*12 out-of-sample period in table 3. When I focus on point prediction accuracy, I rarely find that the BMAs compare favorably with the stochastic volatility benchmark. As evidenced from the table, any relative prediction accuracy gains are very modest and statistically insignificant. In instances, where the no predictability null hypothesis is rejected, it is in favor of the benchmark. Not everything is negative. The BMAs tend to outperform SMA, model selection using BIC and the ridge regression.

With regards to density prediction accuracy, only if I allow for predictive impact through the conditional volatility process do I obtain statistically significant accuracy gains relative to the benchmark. For BMA incl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$, the gains are statistically significant at the 5% level when assigning uniform weights and emphasizing the center of the conditional distribution. BMA incl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ also leads to borderline statistically significant gains at the 10% level when predicting the left tail of the conditional return distribution. Density prediction accuracy gains produced under BMA excl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ relative to the benchmark are statistically significant when assigning uniform weights, emphasizing both tails and focusing on the right/left tail separately. With regards to predicting the left tail, relative gains afforded by BMA excl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$

are statistically significant at the 1% level. At the same time, it must be mentioned that these density accuracy gains are relatively modest, at best around 2%. Similar to previous results, the remaining specifications do not offer any gains. In fact, when the no predictability null hypothesis is rejected, it is rejected in favor of the benchmark model. BMA incl. TVP $\beta_{1:T} \neq 0, \delta_{1:T} = 0$ offers at best gains of less than 0.1%, which due to rounding off is not apparent from the table. The picture is particularly bleak for model selection using BIC. In terms of wCRPS with uniform weights, the benchmark model outperforms this specification by 7%.

For the next three out-of-sample periods, 1965*m*1–2018*m*12, 1976*m*1–2018*m*12 and 1988*m*1–2018*m*12, results reported in tables 4 to 6 display a qualitatively similar pattern. The simple model selection/averaging techniques do not offer statistically significant gains relative to the benchmark. Although BMA specifications, where $\delta_{1:T} = 0$ occasionally offer density accuracy gains, they are at best very modest and not statistically significant. Conversely, BMA incl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ consistently generates statistically significant more accurate density predictions when assigning uniform weights and emphasizing the center of the conditional distribution. It also borderline outperforms the benchmark at the 10% significance level when predicting the left tail. BMA excl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ performs well when assigning

Table 5. Statistical evaluation results over the 1976m1–2018m12 out-of-sample period.

Specification	incl. TVP		excl. TVP		incl. TVP		excl. TVP	
Model	TU	DM	TU	DM	avCRPS	DM	avCRPS	DM
BMA $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.01	0.61	1.00	0.98	1.01	0.46	1.02	0.00
BMA $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	1.00	0.22	0.99	0.19	0.98	0.01	0.99	0.01
BMA $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.00	0.82	1.00	1.00	1.00	0.86	0.99	0.50
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.00	0.73	1.00	0.71	1.00	0.74	1.03	0.00
BMAR $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	1.00	0.43	0.99	0.17	1.01	0.18	0.99	0.33
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.00	0.91	1.00	0.78	1.01	0.28	1.01	0.33
SMA	1.01	0.85	0.99	0.77	1.00	0.76	1.03	0.00
BIC			1.02	0.13			1.03	0.00
Ridge			1.02	0.39			1.00	0.79
	avCRPS-center	DM	avCRPS-center	DM	avCRPS-tails	DM	avCRPS-tails	DM
BMA $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.01	0.47	1.00	0.89	1.00	0.47	1.03	0.00
BMA $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	0.99	0.08	1.00	0.70	1.00	0.67	0.98	0.01
BMA $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.00	0.94	0.99	0.14	1.00	0.82	1.00	0.63
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.00	0.90	1.01	0.28	1.00	0.69	1.03	0.00
BMAR $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	1.00	0.16	1.00	0.79	1.02	0.15	0.99	0.32
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.00	0.83	1.01	0.10	1.01	0.22	1.01	0.45
SMA	1.00	0.76	1.01	0.24	1.00	0.76	1.04	0.00
BIC			1.01	0.05			1.03	0.00
Ridge			1.01	0.23			1.00	0.93
	avCRPS-right tail	DM	avCRPS-right tail	DM	avCRPS-left tail	DM	avCRPS-left tail	DM
BMA $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.01	0.38	1.04	0.00	1.00	0.66	1.01	0.29
BMA $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	1.01	0.46	0.99	0.49	0.99	0.11	0.98	0.08
BMA $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.01	0.51	1.01	0.58	1.00	0.78	0.98	0.36
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.01	0.13	1.03	0.00	1.00	0.61	1.02	0.06
BMAR $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	1.02	0.17	1.00	0.80	1.00	0.50	0.99	0.29
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.01	0.19	1.01	0.32	1.00	0.70	1.01	0.59
SMA	1.01	0.42	1.03	0.00	1.00	0.91	1.03	0.02
BIC			1.04	0.00			1.02	0.02
Ridge			1.00	0.82			1.00	0.84

Note: In the above table, TU (Theil's U) is defined as the MSE produced under the model of interest relative to the MSE produced under the stochastic volatility benchmark. Likewise, avCRPS reports the average (3.2) with the specified weight reported in table 2 produced under the model of interest relative to the avCRPS with the specified weight produced under the stochastic volatility benchmark.

uniform weights and predicting the tails of the conditional distribution. Furthermore, with regards to predicting the left tail of the conditional distribution, gains in accuracy can be as high as 2% relative to the stochastic volatility benchmark and statistically significant at the 5% level.

Results over the remaining out-of-samples, namely, 2000m1–2018m12 and 2010m1–2018m12 reported in tables 7 and 8 are very interesting. With regards to the former, I only obtain statistically significant point prediction accuracy gains in 3 out of the 16 cases. On the other hand, BMA incl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ and BMA excl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ both statistically outperform the benchmark when assigning uniform weights, individually emphasizing the center and tails of the conditional distribution (except the right tail). Here, the predictive gains range between 1% and 3%. The gains are statistically significant at the 5% level for the former while statistically significant at the 1% level for the latter. Furthermore, BMA excl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ generates statistically significant more accurate density predictions relative to the stochastic volatility benchmark at the 1% level when emphasizing the left tail of the conditional distribution with an accuracy gain of 5%. The corresponding gain produced under BMA incl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ is slightly smaller and statistically significant only at the 5% level. Results are also positive for BMAR specifications. For

example, BMAR incl. TVP $\beta_{1:T} \neq 0, \delta_{1:T} = 0$ statistically outperforms the benchmark when emphasizing the center of the conditional distribution. It also performs well when predicting the left tail even though reductions in wCRPS are not statistically significant. Conversely, results from simple prediction combinations, model selection using BIC, Ridge and BMA incl. (excl.) TVP $\beta_{1:T} \neq 0, \delta_{1:T} = 0$ are not positive. These specifications are consistently outperformed by the benchmark.

The last out-of-sample period focuses entirely on the post Great Recession period. In terms of point prediction accuracy, specifications such as BMA excl. TVP $\beta_{1:T} \neq 0, \delta_{1:T} = 0$, SMA and model selection using BIC borderline outperform the benchmark at the 10% significance level. For SMA, reduction in MSE relative to the stochastic volatility benchmark is around 3%. With regards to density prediction accuracy, BMA incl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ and BMA excl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ statistically outperform the stochastic volatility benchmark across all regions of the conditional distribution at the 5% significance level. Results are particularly encouraging when assigning uniform weights and predicting the left tail of the conditional distribution. Here, accuracy gains are statistically significant at the 1% level. The gains themselves range between 3% to 6%. Contrary to previous results, the predictive power afforded from BMA

Table 6. Statistical evaluation results over the 1988m1–2018m12 out-of-sample period.

Specification	incl. TVP		excl. TVP		incl. TVP		excl. TVP	
Model	TU	DM	TU	DM	avCRPS	DM	avCRPS	DM
BMA $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.03	0.30	1.01	0.56	1.01	0.20	1.03	0.00
BMA $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	0.99	0.21	0.99	0.38	0.98	0.01	0.98	0.01
BMA $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.01	0.43	1.01	0.65	1.00	0.85	1.00	0.70
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.00	0.79	1.00	1.00	1.00	0.90	1.03	0.00
BMAR $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	1.00	0.34	0.99	0.34	1.01	0.31	0.99	0.27
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.00	0.93	1.00	0.97	1.01	0.34	1.00	0.73
SMA	1.04	0.51	1.01	0.69	1.01	0.41	1.04	0.00
BIC			1.01	0.30			1.03	0.00
Ridge			1.03	0.33			1.01	0.57
	avCRPS-center	DM	avCRPS-center	DM	avCRPS-tails	DM	avCRPS-tails	DM
BMA $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.01	0.36	1.00	0.98	1.01	0.17	1.04	0.00
BMA $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	0.99	0.05	1.00	0.69	0.99	0.25	0.98	0.01
BMA $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.00	0.94	0.99	0.07	1.00	0.82	1.00	0.89
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.00	0.82	1.00	0.52	1.00	0.80	1.04	0.00
BMAR $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	1.00	0.15	1.00	0.77	1.02	0.26	0.99	0.25
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.00	1.00	1.01	0.18	1.01	0.31	1.00	0.91
SMA	1.01	0.57	1.01	0.29	1.02	0.39	1.05	0.00
BIC			1.01	0.24			1.03	0.00
Ridge			1.02	0.16			1.00	0.80
	avCRPS-right tail	DM	avCRPS-right tail	DM	avCRPS-left tail	DM	avCRPS-left tail	DM
BMA $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.01	0.36	1.04	0.00	1.01	0.15	1.01	0.17
BMA $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	1.00	0.95	0.99	0.44	0.96	0.12	0.95	0.09
BMA $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.00	0.77	1.01	0.56	1.00	0.99	0.98	0.51
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.00	0.43	1.04	0.00	1.00	0.79	1.02	0.16
BMAR $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	1.02	0.23	1.00	0.91	1.00	0.90	0.98	0.29
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.01	0.34	1.01	0.42	1.01	0.59	1.00	0.91
SMA	1.02	0.19	1.04	0.00	1.01	0.70	1.03	0.02
BIC			1.03	0.00			1.02	0.06
Ridge			1.00	0.98			1.01	0.29

Note: In the above table, TU (Theil's U) is defined as the MSE produced under the model of interest relative to the MSE produced under the stochastic volatility benchmark. Likewise, avCRPS reports the average (3.2) with the specified weight reported in table 2 produced under the model of interest relative to the avCRPS with the specified weight produced under the stochastic volatility benchmark.

incl. TVP $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$ relative to the benchmark also tends to be statistically significant. Likewise, results are positive for BMAR incl. TVP specifications when predicting the tails or the center of the conditional distribution. Conversely, BMA excl. TVP $\beta_{1:T} \neq 0, \delta_{1:T} = 0$, SMA, model selection using BIC and Ridge are consistently outperformed by the benchmark.

What can one conclude from out-of-sample results reported in tables 3–8? BMA across a number of predictors while allowing them to impact aggregate equity returns exclusively through the conditional volatility process improves the accuracy of density predictions across almost all out-of-sample periods considered in this study. The biggest payoff occurs when predicting the left tail of the conditional distribution. I believe that this is an important finding because results reported in widely cited studies, such as Dangl and Halling (2012) compare complex models to very simple benchmarks, for example, the historical average estimated using rolling (expanding) OLS, which cannot account for changing data patterns to the same degree as (4.1)–(4.3).[†] Conversely, the suggested framework is able to deliver

statistically significant predictive gains relative to a rather ‘advanced’ benchmark.

Another new and interesting result discovered in this study is that once I allow for the predictive impact to occur via the conditional volatility process, then I essentially do not need to account for time-variation in the model parameters. I reach this conclusion by comparing the performance of BMA excl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ with BMA incl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ across the out-of-sample periods. Here, the former tends to slightly outperform the latter. In this regard, my results are in contrast to studies, such as Dangl and Halling (2012) that consider parameter instability as a major issue. A more detailed investigation regarding this issue is presented in Section 4.3. In a similar fashion, adding parameter instability on top of heteroscedasticity appears to not add much with regards to model performance. On the contrary, it can impact results in a negative way. For example, wCRPS ratios produced under BMA incl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ and BMA excl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ are often similar, whereas the DM p-values for the latter are lower. Evidently, since wCRPS ratios are similar the difference in outcome has to be due to larger $\hat{\sigma}_{HAC}$ in (3.3) for BMA incl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$, which explains the lower DM test statistic and thus the higher p-value. I can interpret the larger $\hat{\sigma}_{HAC}$ for BMA incl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ as a sign of overfitting.

[†] I can effectively make the case by comparing results for BMA incl. TVP $\beta_{1:T} \neq 0, \delta_{1:T} = 0$ with the stochastic volatility benchmark.

Table 7. Statistical evaluation results over the 2000m1–2018m12 out-of-sample period.

Specification	incl. TVP		excl. TVP		incl. TVP		excl. TVP	
Model	TU	DM	TU	DM	avCRPS	DM	avCRPS	DM
BMA $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.04	0.29	1.03	0.38	1.02	0.19	1.03	0.02
BMA $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	0.99	0.24	0.98	0.05	0.98	0.01	0.97	0.00
BMA $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.01	0.64	1.02	0.46	0.99	0.65	0.99	0.54
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.00	0.98	0.98	0.31	1.00	0.68	1.01	0.27
BMAR $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	0.99	0.30	0.98	0.06	1.02	0.32	0.98	0.05
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	0.99	0.74	0.97	0.12	1.00	0.69	0.98	0.24
SMA	1.04	0.63	1.00	0.97	1.01	0.71	1.03	0.06
BIC			1.00	0.90			1.02	0.11
Ridge			1.04	0.44			1.01	0.57
	avCRPS-center	DM	avCRPS-center	DM	avCRPS-tails	DM	avCRPS-tails	DM
BMA $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.01	0.34	1.00	0.79	1.02	0.16	1.04	0.01
BMA $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	0.97	0.06	0.98	0.00	0.96	0.03	0.95	0.00
BMA $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	0.99	0.12	0.98	0.08	1.00	0.80	0.99	0.69
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	0.99	0.28	0.99	0.44	1.00	0.89	1.02	0.15
BMAR $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	0.99	0.05	0.99	0.01	1.03	0.27	0.97	0.08
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	0.99	0.08	0.99	0.25	1.01	0.52	0.98	0.27
SMA	1.00	0.99	1.00	0.93	1.01	0.64	1.04	0.03
BIC			1.00	0.90			1.02	0.06
Ridge			1.03	0.18			1.00	0.79
	avCRPS-right tail	DM	avCRPS-right tail	DM	avCRPS-left tail	DM	avCRPS-left tail	DM
BMA $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.02	0.20	1.07	0.00	1.02	0.23	1.00	0.96
BMA $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	1.00	0.89	0.98	0.44	0.96	0.08	0.95	0.02
BMA $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.01	0.62	1.02	0.39	0.98	0.42	0.97	0.31
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.01	0.49	1.04	0.01	0.99	0.42	0.99	0.55
BMAR $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	1.04	0.20	1.00	0.84	1.00	0.94	0.96	0.14
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.01	0.50	1.00	0.85	1.00	1.00	0.97	0.24
SMA	1.04	0.14	1.06	0.00	0.99	0.70	1.00	0.99
BIC			1.04	0.02			1.00	0.99
Ridge			1.00	0.97			1.02	0.29

Note: In the above table, TU (Theil's U) is defined as the MSE produced under the model of interest relative to the MSE produced under the stochastic volatility benchmark. Likewise, avCRPS reports the average (3.2) with the specified weight reported in table 2 produced under the model of interest relative to the avCRPS with the specified weight produced under the stochastic volatility benchmark.

4.2. Prediction performance and the business cycle

Exploring the connections between the business cycle and relative out-of-sample performance may further illustrate the sources of predictive power afforded by the most successful BMAs relative to the stochastic volatility benchmark and other specifications. Therefore, let ΔL_s denote the difference between the density prediction produced under the stochastic volatility model (labeled as *b*) relative to model of interest (labeled as *l*).

Panel (a) of figure 2 displays the cumulative $\Delta L_s = wCRPS_s^{(b)} - wCRPS_s^{(l)}$ using $w(z) = 1$ for (4.1)–(4.3) relative to BMA incl. TVP $\beta_{1:T} \neq 0, \delta_{1:T} = 0$.[†] Here, periods where the line slopes upward represent periods in which BMA incl. TVP $\beta_{1:T} \neq 0, \delta_{1:T} = 0$ outperforms the benchmark, while downward-sloping segments indicate periods in which the benchmark density prediction is more accurate. Vertical lines denote NBER business cycle dates. Under the null hypothesis of equal predictive ability, the plot ought to trend steadily downward, as additional estimation error associated with the more heavily parameterized BMA incl. TVP $\beta_{1:T} \neq 0, \delta_{1:T} = 0$ model decreases density prediction accuracy relative to the benchmark. As displayed in panel (a), the relative predictive

power afforded by BMA incl. TVP $\beta_{1:T} \neq 0, \delta_{1:T} = 0$ is short-lived, prone to instability and mainly tends to concentrate on the onset of recessions. In other periods, the cumulative ΔL_s line is either flat or slightly decreasing. For example, I observe short-lived spikes in ΔL_s in favor of BMA incl. TVP $\beta_{1:T} \neq 0, \delta_{1:T} = 0$ following a series of recessions in the 1970s and 1980s. It also delivers density accuracy gains on the onset of the Great Recession. However, any such predictive gains are more than offset by losses in other periods, which in turn, explains why the avCRPS ratios are consistently close or slightly above one in tables 3–8.

The corresponding plot for BMA incl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ is displayed in panel (b) of figure 2. As evidenced from the panel, allowing the predictors to impact aggregate equity returns exclusively through the conditional volatility process results in better out-of-sample performance compared to panel (a), particularly from the early 2000s till the end of the sample. Here, the cumulative ΔL_s line consistently slopes upwards. Contrary to panel (a), the upward spikes in ΔL_s on the onset of recessions are more notable. Likewise, BMA incl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ tends to afford sizable accuracy gains as the economy recovers from a recession. Not everything is positive though. We still observe periods, where the cumulative ΔL_s line slopes downwards. Good illustrations are the period running from the 1960s through 1980s and from 1997 through 2000.

[†] The conclusions are qualitatively similar when emphasizing different areas of the conditional distribution or using point predictions.

Table 8. Statistical evaluation results over the 2010m1–2018m12 out-of-sample period.

Specification	incl. TVP		excl. TVP		incl. TVP		excl. TVP	
Model	TU	DM	TU	DM	avCRPS	DM	avCRPS	DM
BMA $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.02	0.21	0.98	0.10	1.01	0.06	1.02	0.14
BMA $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	0.99	0.35	0.98	0.11	0.96	0.00	0.96	0.00
BMA $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	0.99	0.40	0.98	0.58	0.96	0.00	0.97	0.05
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.00	0.87	0.97	0.13	0.99	0.61	1.02	0.20
BMAR $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	0.99	0.27	0.98	0.12	0.99	0.13	0.99	0.41
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	0.99	0.67	0.98	0.25	0.99	0.18	0.99	0.75
SMA	1.02	0.29	0.97	0.11	1.01	0.41	1.03	0.06
BIC			0.98	0.10			1.02	0.23
Ridge			1.04	0.17			1.00	0.91
	avCRPS-center	DM	avCRPS-center	DM	avCRPS-tails	DM	avCRPS-tails	DM
BMA $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.01	0.35	0.99	0.31	1.01	0.03	1.03	0.06
BMA $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	0.99	0.05	0.99	0.02	0.95	0.00	0.95	0.00
BMA $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	0.99	0.20	0.98	0.26	0.94	0.00	0.96	0.04
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	0.98	0.26	0.99	0.28	1.00	1.00	1.03	0.08
BMAR $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	0.99	0.02	0.99	0.06	0.99	0.23	0.99	0.52
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	0.98	0.08	0.99	0.46	0.99	0.32	1.00	0.83
SMA	0.99	0.62	0.99	0.38	1.01	0.12	1.05	0.02
BIC			1.00	0.93			1.02	0.16
Ridge			1.01	0.53			0.99	0.75
	avCRPS-right tail	DM	avCRPS-right tail	DM	avCRPS-left tail	DM	avCRPS-left tail	DM
BMA $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.01	0.29	1.05	0.00	1.02	0.10	1.00	0.92
BMA $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	0.98	0.16	0.97	0.02	0.95	0.00	0.94	0.00
BMA $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	0.97	0.03	0.99	0.67	0.94	0.00	0.94	0.02
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.01	0.27	1.05	0.00	0.98	0.10	0.99	0.54
BMAR $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	1.01	0.17	1.02	0.16	0.97	0.01	0.96	0.09
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.00	0.51	1.02	0.07	0.97	0.07	0.97	0.27
SMA	1.03	0.01	1.05	0.00	0.98	0.22	1.01	0.70
BIC			1.03	0.00			1.00	0.86
Ridge			1.01	0.58			0.98	0.48

Note: In the above table, TU (Theil's U) is defined as the MSE produced under the model of interest relative to the MSE produced under the stochastic volatility benchmark. Likewise, avCRPS reports the average (3.2) with the specified weight reported in table 2 produced under the model of interest relative to the avCRPS with the specified weight produced under the stochastic volatility benchmark.

In panel (c), I compare the performance of BMA excl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ with the benchmark. Similar to results for BMA incl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$, we observe a degree of affinity between out-of-sample performance and the business cycle. Likewise, as illustrated by the slope of the cumulative ΔL_s line, BMA excl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ performs very well from the early 2000s till the end of the sample. However, contrary to BMA incl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$, one does not observe prolonged periods, where the cumulative ΔL_s line slopes downwards. In fact, a closer look at the behavior of BMA incl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ relative to BMA excl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ clearly demonstrates that density predictions from the former tend to vary substantially, sometimes performing well and sometimes really bad. This in turn, can help explain why it fails to outperform BMA excl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$. In other words, results from panels (b) and (c) suggest that BMA incl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ may be suffering from overfitting (i.e. the model becomes too flexible). According to results displayed in panels (b) and (c), once the predictive impact of the economic variables is accounted for through the conditional volatility process, then there is apparently no need to allow for time-variation in the model parameters.

The above pattern for BMA excl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ holds regardless, which region of the conditional distribution

is emphasized. For example, in panel (d), I display cumulative ΔL_s for BMA excl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$, where the left tail of the conditional distribution is emphasized. Here, I observe similar patterns as those reported in panel (c). Furthermore, the model performs particularly well on the onset of the Great Recession.

To supplement the graphical analysis presented in figure 2, I also examine regressions of ΔL_s on the growth rate in quarterly real U.S. GDP (gdp) over the out-of-sample periods. In other words,

$$\Delta L_s = a + b \cdot gdp + e_s, \quad e_s \sim N(0, \sigma_e^2). \quad (4.4)$$

In the above regression, a negative estimate of b indicates a countercyclical pattern in predictive power relative to the stochastic volatility benchmark. After estimating (4.4), I observe that $\hat{b}$ is negative and statistically significant mainly for BMA excl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ and to a lesser degree for BMA incl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$. For instance, for the former model, I estimate b at -0.12 with a t -statistic of -2.08 over 1947m1–2018m12. For the 2000m1–2018m12 out-of-sample period, $\hat{b}$ is -0.27 with a t -statistic of -2.45 . Conversely, for a number of specifications, evidence of countercyclical pattern in predictive power is relatively weaker as the null hypothesis, $b = 0$, cannot be rejected. In fact,

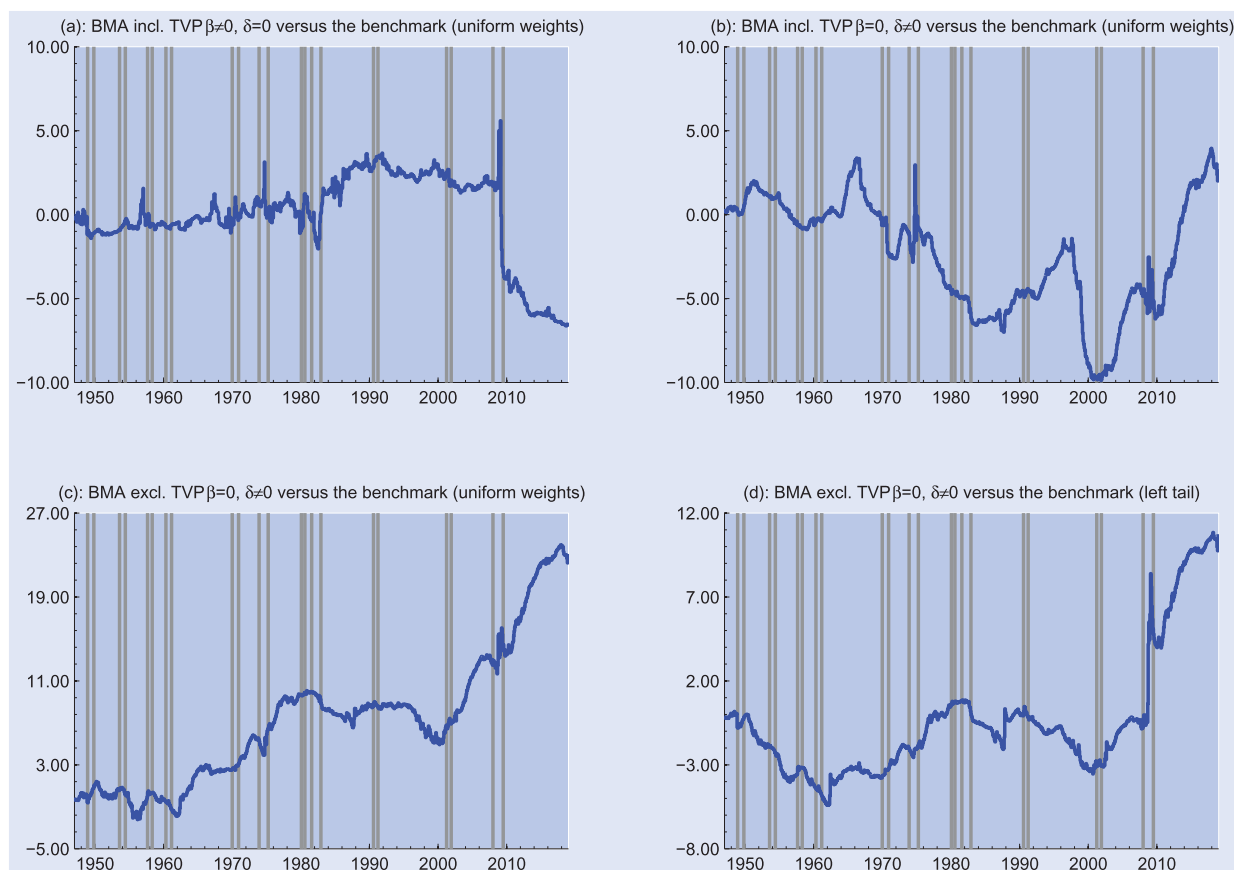


Figure 2. Out-of-sample density prediction performance of different BMA models and the business cycle.

These panels display the out-of-sample cumulative $\Delta L_s = wCRPS_s^{(b)} - wCRPS_s^{(l)}$ given the weight of interest produced under the benchmark (labeled as b) and the model of interest (labeled as l). In panels (a) to (d), vertical lines indicate economic recessions using National Bureau of Economic Research (NBER) business cycle dating.

for certain models, such as BMA incl (excl). TVP $\beta_{1:T} \neq 0$, $\delta_{1:T} = 0$ predictive power appears to be slightly procyclical. Overall, the graphical analysis in figure 2 and the simple regression analysis above help explain the success of the suggested BMA specification.

4.3. Characterization of the BMA models

The previous section provided some interesting results regarding the relationship between out-of-sample prediction performance and the business cycle. However, given that the BMA models are fairly sophisticated and contain a number of different features, I provide more information regarding their characteristics in this section.

The first analysis focuses on what occurs when different model features are turned on or off. I also relate my results to the main findings in Dangl and Halling (2012). Particularly, using BMAs that assume constant conditional innovation volatility, the authors find evidence in favor of BMA incl. TVP relative to BMA excl. TVP in the context of predicting aggregate equity returns. Panel (a) of figure 3 displays the cumulative $\Delta L_s = wCRPS_s^{(c)} - wCRPS_s^{(l)}$ using $w(z) = 1$, between BMA excl. TVP $\beta_{1:T} \neq 0$, $\delta_{1:T} = 0$ (indexed as c) and BMA incl. TVP $\beta_{1:T} \neq 0$, $\delta_{1:T} = 0$, where $\kappa_{1:T} = 1$ (indexed as l). Accordingly, both BMAs assume constant conditional innovation volatility. However, the latter allows for

time-variation in the regression coefficients, β_t . In panel (a), a positive slope of the cumulative ΔL_s line signifies that BMA incl. TVP $\beta_{1:T} \neq 0$, $\delta_{1:T} = 0$, where $\kappa_{1:T} = 1$ outperforms BMA excl. TVP $\beta_{1:T} \neq 0$, $\delta_{1:T} = 0$ and vice versa. As evidenced from panel (a), similar to Dangl and Halling (2012) results are generally positive for the former. In panel (b), I compare BMA incl. TVP $\beta_{1:T} \neq 0$, $\delta_{1:T} = 0$ (indexed as l) with BMA incl. TVP $\beta_{1:T} \neq 0$, $\delta_{1:T} = 0$, where $\lambda_{1:T} = 1$ (indexed as c) to evaluate if there is still evidence in favor of time-variation in β_t once I also allow for time-varying conditional innovation volatility. Here, except for short-lived spikes on the onset of certain recessions, the slope of the cumulative ΔL_s line is largely negative. This in turn, implies that the evidence in favor of BMA incl. TVP with time-invariant volatility established in Dangl and Halling (2012) is mainly driven by the constant volatility assumption. In other words, if the evolution in conditional innovation volatility is not modeled appropriately (which is the case if one assumes constant volatility for equity return data despite evidence of heteroscedasticity, see for example, figure 1), then the omitted part of model complexity is reflected in a different part of the model, namely, time-varying regression coefficients.

In panels (c) and (d), I repeat the above analysis. However, I also simultaneously allow the predictors to impact the dependent variable through the conditional volatility channel. Here, results are qualitatively similar to those in panels

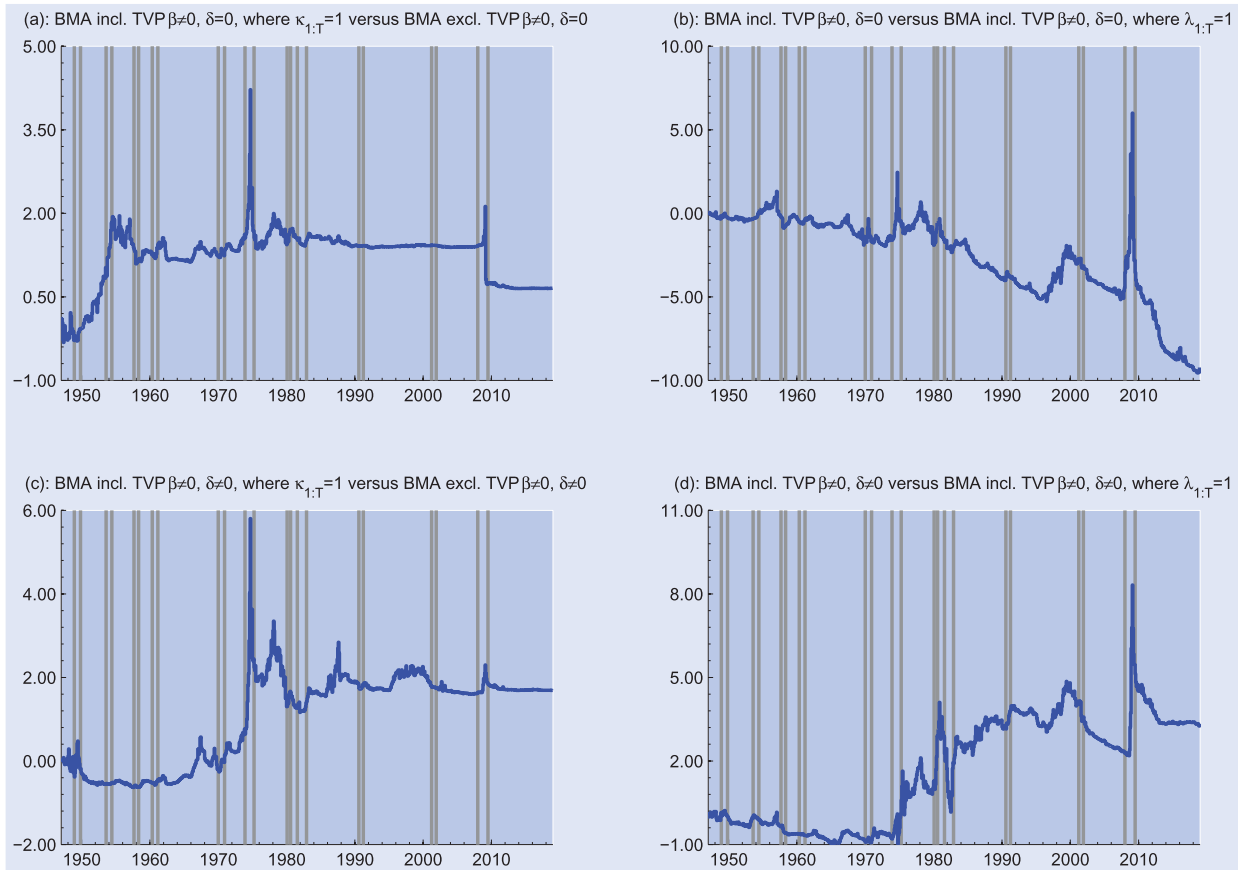


Figure 3. Disentangling the effects of different model features over the out-of-sample.

These panels display the out-of-sample cumulative $\Delta L_s = wCRPS_s^{(c)} - wCRPS_s^{(l)}$ given uniform weights, $w(z) = 1$, produced under various models. In panels (a) and (c), a positive slope signifies evidence in favor of BMA incl. TVP $\beta_{1:T} \neq 0, \delta_{1:T} = 0$, where $\kappa_{1:T} = 1$ and BMA incl. TVP $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$, where $\kappa_{1:T} = 1$, respectively. In panels (b) and (d), a positive slope signifies evidence in favor of BMA incl. TVP $\beta_{1:T} \neq 0, \delta_{1:T} = 0$ and BMA incl. TVP $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$, respectively. In panels (a) to (d), vertical lines indicate economic recessions using National Bureau of Economic Research (NBER) business cycle dating.

(a) and (b). As displayed in panel (c), BMA incl. TVP $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$ leads to gains compared to BMA incl. TVP $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$, where $\kappa_{1:T} = 1$, especially during the mid 1970s. Furthermore, the pattern of gains is somewhat similar to those in panel (a). Conversely, assuming $\lambda_{1:T} \neq 1$ and $\kappa_{1:T} \neq 1$ compared to $\lambda_{1:T} = 1$ and $\kappa_{1:T} \neq 1$ for BMA, where $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$ results in a more volatile cumulative ΔL_s line. Similar to panel (b), the relatively more frequent downward-sloping segments indicate that allowing for time-variation in the regression coefficients once $\kappa_{1:T} \neq 1$ tends to lead to performance deterioration due to the too heavy parameterization.

Results in the previous section establish that BMA incl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ and BMA excl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ tend to perform better than their competitors. This section sheds more light on why this is the case. I start by comparing the average number of predictors across different BMAs over time, namely, (2.9). In panel (a) of figure 4, I report (2.9) for BMA incl. TVP $\beta_{1:T} \neq 0, \delta_{1:T} = 0$. Interestingly, although there are 12 predictors and thus more than 4000 different predictor combinations, the expected model size is at most around two. Furthermore, despite a short-lived upward spike in (2.9) during the Great Recession, I observe a decreasing trend in expected model size. On the other hand, as displayed in panel (b), BMA incl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ (with

qualitatively similar results for BMA excl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$) contains on average a higher number of predictors throughout the out-of-sample period. It is around 4 from the start of the out-of-sample until the early 2000s. Thereafter, one observes an upward shift and (2.9) varies between 5 and 7. I also observe a degree of affinity between the average model size and the business cycle: Average size tends to increase on the onset of a recession and then decreases following the recovery. At the same time, I observe a degree of heterogeneity in this regard. For example, (2.9) responds more aggressively with respect to recessions in the early 2000s and 2008 than other periods.

Overall, results from panels (a) and (b) demonstrate why BMA, where $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ outperforms BMA, where $\beta_{1:T} \neq 0, \delta_{1:T} = 0$. The former contains on average more predictors, which provide information useful for improving prediction accuracy. As indicated by (2.9), predictors that perform well only on the onset or during recessions enter and exist the model at the right time, which further helps explain why the former outperforms the latter, namely, it is very good at taking advantage of the predictive power afforded by the economic predictors while at the same time avoiding overfitting. Using the same logic, one can intuitively explain why BMA excl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ tends to perform marginally better than BMA incl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$. As

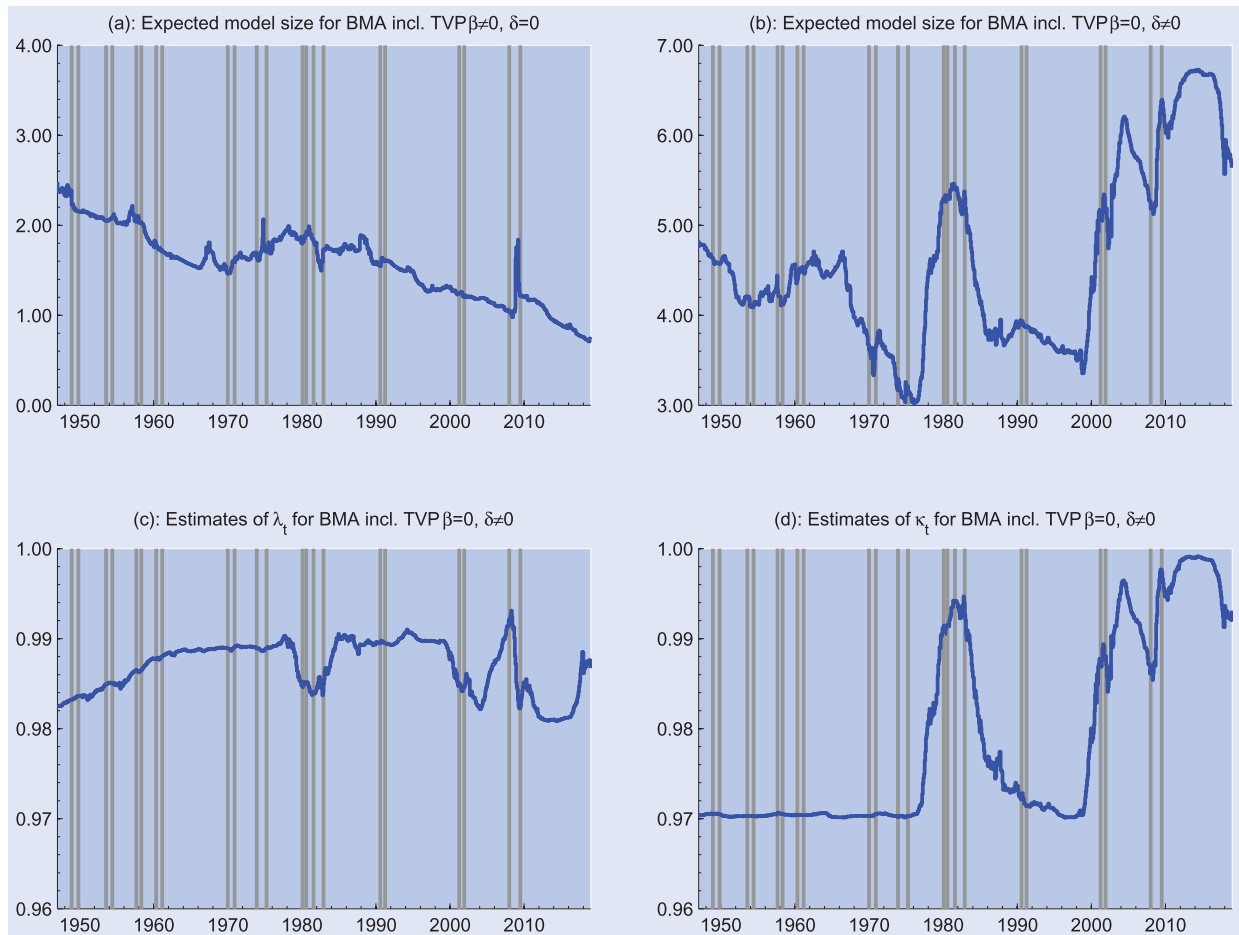


Figure 4. Expected model size and the degree of parameter instability for different BMAs.

Panels (a) and (b) display the average model size, (2.9), over the 1947m1–2018m12 out-of-sample period for the model of interest. Panels (c) and (d) display the model average estimates of λ and κ over the same out-of-sample period. In panels (a) to (d), vertical lines indicate economic recessions using National Bureau of Economic Research (NBER) business cycle dating.

mentioned earlier, the expected model size is similar for BMA excl. TVP $\beta_{1:T} = 0$, $\delta_{1:T} \neq 0$ and BMA incl. TVP $\beta_{1:T} = 0$, $\delta_{1:T} \neq 0$. Hence, the better relative performance of the former must be due to overfitting caused by the incl. TVP extension of the latter. In other words, allowing for time-variation in the model parameters makes the model too flexible. Consequently, any additional gains in prediction accuracy afforded by allowing model parameters to vary over time is more than offset by increases in prediction variance relative to the excl. TVP specification.

Next, I evaluate the importance of individual predictors in BMA incl. TVP $\beta_{1:T} = 0$, $\delta_{1:T} \neq 0$ and BMA excl. TVP $\beta_{1:T} = 0$, $\delta_{1:T} \neq 0$. For each predictor in table 1, I use the sum of probabilities across all models that include the variable of interest as the measure of importance, namely, $\sum_{i=1}^k 1_{(i \in l)} p(\mathcal{M}_i | y_{1:t})$, $l = 1, \dots, p$. This in turn, allows me to evaluate how much support each predictor receives from the data. Table 9 presents inclusion probabilities at six points in time: 1947m1, 1965m1, 1976m1, 1988m1, 2000m1 and 2010m1. Results for BMA incl. TVP $\beta_{1:T} = 0$, $\delta_{1:T} \neq 0$ are reported on the left side of table 9. Here, the dividend yield (dy) and the default return spread (dfr) consistently receive high inclusions probabilities across the selected dates. On the other hand, variables, such as earnings-to-price ratio (ep), dividend-payout ratio (dpayr), book-to-market ratio (bmr) and

S&P 500 monthly log-realized volatility (svar) obtain high inclusion probabilities for 2000m1 and 2010m1. I also observe that a number of predictors consistently obtain probabilities around 50% (i.e. the unconditional prior value). For example, long-term government bond yields (lty), long-term return on government bonds (ltr), default yield spread (dfy) and inflation rate (inf) rarely obtain inclusion probabilities greater than 50%. On the right side of table 9, I report results for BMA excl. TVP $\beta_{1:T} = 0$, $\delta_{1:T} \neq 0$. Here, dpayr and svar receive inclusion probabilities that exceed 90% at all points in time. Conversely, I observe a degree of switching for ep and bmr. Finally, similar to BMA incl. TVP $\beta_{1:T} = 0$, $\delta_{1:T} \neq 0$, a number of variables receive inclusion probabilities close to 50%.

In panels (c) and (d) of figure 4, I display the model average estimates of λ_t and κ_t produced under BMA incl. TVP $\beta_{1:T} = 0$, $\delta_{1:T} \neq 0$. Similar to Dangi and Halling (2012) estimates of λ_t vary between 0.98 and 0.99 throughout most of the out-of-sample period. Likewise, for a large portion of the out-of-sample period, one does not observe a degree of affinity between the estimate of λ_t and the business cycle. In fact, λ_t gradually increases from 0.98 to 0.99 independent of the business cycle. Towards the end of the sample, one tends to observe a link between λ_t and the business cycle: λ_t decreases as one approaches a recession and subsequently increases

Table 9. Inclusion probabilities of individual predictors.

Period	1947m1	1965m1	1976m1	1988m1	2000m1	2010m1	1947m1	1965m1	1976m1	1988m1	2000m1	2010m1
Specification	BMA incl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$						BMA excl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$					
Dy	0.97	0.96	0.97	0.81	0.69	0.19	0.25	0.26	0.28	0.15	0.22	0.15
Ep	0.15	0.23	0.05	0.15	0.21	0.69	0.27	0.56	0.79	0.51	0.91	0.93
Dpayr	0.03	0.04	0.03	0.20	0.37	0.98	0.80	0.89	0.92	0.99	0.99	1.00
Bmr	0.15	0.23	0.05	0.37	0.66	0.63	0.34	0.61	0.81	0.44	0.88	0.86
Svar	0.01	0.00	0.02	0.20	0.37	0.99	0.93	1.00	1.00	1.00	1.00	1.00
Ntis	0.19	0.10	0.03	0.06	0.14	0.52	0.49	0.50	0.45	0.50	0.51	0.64
Tbl	0.48	0.57	0.16	0.08	0.14	0.29	0.51	0.45	0.43	0.10	0.07	0.11
Lty	0.46	0.65	0.06	0.01	0.02	0.06	0.51	0.48	0.43	0.12	0.08	0.07
Ltr	0.50	0.20	0.11	0.06	0.09	0.28	0.48	0.49	0.47	0.40	0.37	0.37
Dfr	0.96	0.99	0.97	0.85	0.74	0.36	0.52	0.53	0.52	0.49	0.48	0.46
Dfy	0.48	0.25	0.39	0.50	0.40	0.53	0.50	0.50	0.51	0.55	0.53	0.57
Inf	0.41	0.23	0.16	0.42	0.41	0.56	0.50	0.50	0.50	0.50	0.50	0.50

Note: Columns two to seven display predictor inclusion probabilities for BMA incl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$. Columns eight to thirteen display predictor inclusion probabilities for BMA excl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$. I refer the reader to table 1 for the definition of the variables and their abbreviations.

following the recovery. I observe a similar pattern for $\kappa_{1:T}$. Initially, I estimate the discount factor at 0.97 per month. After a short-lived increase during the early 1980s, κ_t reverts back to its prior levels. Similar to results reported in panel (c), I observe a link between the estimate of κ_t the business cycle towards the end of the sample.

Finally, I follow Dangl and Halling (2012), Byrne *et al.* (2018) and perform a variance decomposition analysis. Similar to the authors, I decompose prediction uncertainty into the observational variance, estimation uncertainty in the coefficient vectors, uncertainty with respect to the choice of the predictors and uncertainty with respect to time-variation in the model parameters. Overall, I find that results are in large qualitatively similar to those reported in Dangl and Halling (2012). The dominant source of uncertainty is the observational variance. Compared to BMAs, where $\beta_{1:T} \neq 0, \delta_{1:T} = 0$, I observe that the observational variance's share of the total variance is higher for BMAs, where $\delta_{1:T} \neq 0$. However, this is expected as for the latter, the observational variance is a function of the log-volatility process as well as the predictors. On the other hand, expected variance from errors in the estimation of the coefficient vectors is lower for the latter specifications. The same pattern is observed in the context of uncertainty with respect to the choice of the predictors. Lastly, BMA incl. TVP models display a similar degree of uncertainty with regards to time-variation in the model parameters. More precisely, uncertainty about the correct degree of time-variation in the model parameters is relatively low except for individual periods, such as the end of the 1980s, the beginning of the 1990s and the Great Recession.

4.4. Economic evaluation

As the final analysis, I evaluate whether the identified levels of statistical gains discovered in Section 4.1 can be translated to economic gains. Particularly, following the framework discussed in Section 3.3, I evaluate the gains in utility for an investor who uses conditional mean and volatility predictions produced under the predictive models relative to the benchmark. I analyze the gains in utility under two risk aversion parameters, namely, $\gamma = 3$ and 6.

Table 10 reports the average gains in utility produced under the predictive models minus the average gain in utility produced under the benchmark across the considered out-of-sample periods. Here, I observe that BMA that allows the predictors to impact aggregate equity returns through the conditional mean as well as the volatility process results in the most consistent pattern of gains relative to the stochastic volatility benchmark. For this BMA model, relative utility gains are between 1% to 2% over the first four out-of-sample periods, whereas for the remaining out-of-sample periods, one obtains relative average gains as high as 4%. Somewhat similar to results reported in Section 4.1, I do not find firm evidence in favor of BMA incl. TVP $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$ relative to BMA excl. TVP $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$ or vice versa: The former specification performs marginally better over the initial out-of-sample periods, whereas the picture is reversed over 2000m1–2018m12 and 2010m1–2018m12.

More interesting results are observed when comparing BMA incl. (excl.) TVP $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$ with other predictive models. For example, comparing the performance of BMA incl. (excl.) TVP $\beta_{1:T} \neq 0, \delta_{1:T} = 0$ with BMA incl. (excl.) TVP $\beta \neq 0, \delta_{1:T} \neq 0$, I observe that the latter performs notably better over the initial out-of-sample periods, whereas over 2000m1–2018m12 and 2010m1–2018m12, the utility gains are of minor magnitude. I also observe very good results for BMAR incl. (excl.) TVP $\beta_{1:T} \neq 0, \delta_{1:T} = 0$ and BMAR incl. (excl.) TVP $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$. Likewise, SMA, model selection using BIC and Ridge perform well relative to BMA incl. (excl.) TVP $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$ in certain instances, see table 10. However, turning off the predictive mean channel and allowing the predictive impact to occur only through the conditional volatility process reduces utility gains by as much as 50% relative to BMA incl. (excl.) TVP $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$.

What explains the difference between conclusions for BMA incl. (excl.) TVP $\beta_{1:T} \neq 0, \delta_{1:T} = 0$ and BMA incl. (excl.) TVP $\beta \neq 0, \delta_{1:T} \neq 0$ in this section with those reported in Section 4.1. To begin with, it is important to make the reader aware that wCRPS takes into account the whole shape of the conditional return distribution, whereas with regards to utility comparison, ϖ^* in (3.4) is only the function of the first

Table 10. Economic evaluation over the considered out-of-sample periods.

Out-of-sample period	1947m1–2018m12		1965m1–2018m12		1976m1–2018m12		1988m1–2018m12		2000m1–2018m12		2010m1–2018m12	
Specification	incl. TVP	excl. TVP	incl. TVP	excl. TVP	incl. TVP	excl. TVP	incl. TVP	excl. TVP	incl. TVP	excl. TVP	incl. TVP	excl. TVP
$\gamma = 3$												
BMA $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	0.98	0.46	1.06	0.85	1.25	1.07	1.06	1.44	0.95	2.20	0.45	1.37
BMA $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	0.09	−0.54	0.12	−0.19	0.16	−0.23	0.22	−0.33	0.49	−0.14	1.44	0.17
BMA $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.81	1.04	1.65	1.11	1.97	1.39	2.12	1.92	2.44	3.11	1.87	4.17
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	1.14	2.23	0.87	1.71	1.01	1.52	1.18	1.74	1.84	2.54	3.71	3.17
BMAR $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	0.06	−0.53	0.13	−0.16	0.16	−0.21	0.23	−0.29	0.36	−0.08	0.71	0.30
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	0.92	1.80	0.86	1.10	0.95	0.83	1.02	0.90	1.29	1.50	2.43	2.42
SMA	1.08	0.97	0.70	0.67	0.78	0.84	0.29	1.10	0.84	2.11	1.25	3.15
BIC		0.86		0.67		0.18		0.19		0.53		−0.07
Ridge		0.72		1.03		0.32		0.25		1.63		1.71
$\gamma = 6$												
BMA $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	0.57	0.23	0.59	0.43	0.70	0.54	0.55	0.72	0.50	1.11	0.22	0.69
BMA $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	0.04	−0.27	0.06	−0.09	0.08	−0.12	0.12	−0.16	0.25	−0.07	0.74	0.09
BMA $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	1.16	0.76	1.00	0.66	1.17	0.82	1.26	1.14	1.53	1.85	1.07	2.27
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} = 0$	0.90	1.46	0.65	1.08	0.77	0.93	0.95	1.04	1.51	1.43	3.11	1.66
BMAR $\beta_{1:T} = 0, \delta_{1:T} \neq 0$	0.03	−0.27	0.07	−0.08	0.08	−0.10	0.11	−0.14	0.18	−0.04	0.35	0.15
BMAR $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$	0.72	1.33	0.60	0.74	0.67	0.50	0.72	0.53	0.94	0.83	1.79	1.32
SMA	0.74	0.63	0.43	0.37	0.48	0.46	0.17	0.61	0.47	1.15	0.65	1.74
BIC		0.65		0.40		0.12		0.14		0.33		−0.03
Ridge		0.60		0.80		0.21		0.19		1.06		0.90

Note: Results in the above table are produced assuming an investor with mean-variance preference and a relative risk aversion, γ , equal to 3 and 6, respectively. Furthermore, I limit the share invested into the S&P 500 index to be between 0% and 150%.

and second moments of the conditional distribution. Hence, it does not contain information about the higher moments. This in turn, helps explain the difference in conclusions to some degree. Furthermore, I look more carefully at the weight attached to the risky asset over the out-of-sample period. What is immediately apparent is that the weight attached to the risky asset changes more frequently for BMA incl. (excl.) TVP $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$ compared to other specifications. In certain instances, the number of switches is twice as high. Furthermore, these frequent changes also tend to be associated with increases in utility (although the magnitude of the increases are often marginal and vary over time). This again helps explain the better relative performance of BMA incl. (excl.) TVP $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$ to some degree. What is even more interesting is the fact that contrary to BMA, where only one source of predictability is turned on, an investor using predictions from BMA incl. (excl.) TVP $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$ tends to time the market much better. In fact, this is where I find the main explanation for the superior performance of BMA incl. (excl.) TVP $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$ relative to BMA incl. (excl.) TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$. Particularly, as evidenced by ϖ^* over the out-of-sample period, an investor using predictions produced under BMA incl. (excl.) TVP $\beta_{1:T} \neq 0, \delta_{1:T} \neq 0$ tends to withdraw from the market more quickly on the onset of a recession. Likewise, she times moving back to the market following the recovery phase reasonably well. In contrast, an investor using predictions produced under other specifications is either overly optimistic, and thus fails to withdraw from the market at the right time as the economy enters a recession, or is too slow to move back into the market as the economy recovers. I find that BMA incl. (excl.) TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ seems to suffer mainly from the latter.

Finally, although the BMA-based results in this study cannot be directly compared with studies, such as Beckmann and Schüssler (2014), Pettenuzzo and Ravazzolo (2016), one observes certain similarities. For example, the latter study finds that time-varying parameter models with SV effects perform very well in terms of generating accurate density predictions. The positive density prediction results also translate to economic gains. Beckmann and Schüssler (2014) reach a similar conclusion with respect to economic gains but density prediction results are less positive.[†] Accordingly, neither studies account for the predictive impact through the conditional volatility channel. However, if I only consider models, where the predictive impact occurs via the conditional mean channel, then I find commonalities with results reported in Pettenuzzo and Ravazzolo (2016).

5. Conclusion and future research

This study suggests a BMA framework, where besides addressing parameter instability and model uncertainty a new model feature is introduced, namely, predictors can impact the

dependent variable through the conditional volatility process. With regards to density prediction accuracy, I find statistically significant predictive gains across most out-of-sample periods for BMAs, where predictors impact aggregate equity returns only through the conditional volatility process. For instance, BMA excl. TVP $\beta_{1:T} = 0, \delta_{1:T} \neq 0$ performs particularly well when predicting the left tail of the conditional return distribution. Conversely, other specifications often fail to outperform the stochastic volatility benchmark. I explore these findings by evaluating predictive performance across the business cycle and examining average model size over time. Finally, I find that the BMA, where predictors impact aggregate equity returns through the conditional mean as well as the volatility process leads to the highest gains in utility relative to other specifications and the stochastic volatility benchmark across most out-of-sample periods.

Finally, the econometric framework in this study is univariate, in the sense that there is only one dependent variable. By relying on BMA as well as useful approximations via discount factors, this study provides a sophisticated but parsimonious framework that can be applied without resorting to running simulations. Evidently, extending this framework to a multivariate setting would be an interesting step forward. At the same time, from a computational viewpoint, this extension is difficult. Specifically, because of the more complex model structure, I believe that relying on approximations via discount factors would not be possible. Therefore, one would be highly likely need to resort to running simulations. With regard to the latter, it is relatively easy to go from the univariate to the multivariate setting for a single model. Here, one can use the conditioning features of various MCMC samplers, such as Gibbs sampling combined with data augmentation, and sequentially generate draws from the conditional posterior distributions of the latent states and model parameters. Likewise, besides the conditional volatility channel, one can allow the predictors to impact the conditional innovation covariances between the dependent variables. However, with regard to performing recursive estimation of a model for out-of-sample prediction, one faces a huge computational challenge. This gets even more complicated if one wishes to perform BMA and thus have to estimate many individual models to perform averaging later on. This challenge can be addressed using brute force if one has only a few predictors at hand. Alternatively, one can rely on graphical process unit (GPU) parallel computing, which in turn, speed-ups computation by up to seventy times compared to a standard desktop. Here, the recent study by Casarin *et al.* (2015) provides a very good introduction and opens possible avenues for future research.

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[†] With respect to density prediction performance, Beckmann and Schüssler (2014) rely on the log-score criterion. However, as pointed out in Gneiting and Ranjan (2011), among others, log-score does not reward values from the predictive density that are close but not equal to the actual realized value. It is also very sensitive to outliers. Conversely, wCRPS does not suffer from these issues.

References

- Alquist, R., Kilian, L. and Vigfusson, R.J., Forecasting the price of oil. In *Handbook of Economic Forecasting*, edited by A. Timmermann and G. Elliott, 2013 (Elsevier: Amsterdam).
- Ang, A. and Bekaert, G., Stock return predictability: Is it there? *Rev. Financ. Stud.*, 2007, **20**, 651–707.
- Avramov, D., Stock return predictability and model uncertainty. *J. Financ. Econ.*, 2002, **64**, 423–458.
- Bansal, R. and Yaron, A., Risks for the long run: A potential resolution of asset pricing puzzles. *J. Finance*, 2004, **59**, 1481–1509.
- Beckmann, J. and Schüssler, R., Forecasting equity premia using bayesian dynamic model averaging discussion paper, Center for Quantitative Economics (CQE), University of Muenster, 2014.
- Brennan, M.J. and Xia, Y., Stock price volatility and equity premium. *J. Monet. Econ.*, 2001, **47**, 249–283.
- Brogard, J. and Detzel, A., The asset-pricing implications of government economic policy uncertainty. *Manage. Sci.*, 2015, **61**, 3–18.
- Brunnermeier, M.K. and Pedersen, L.H., Market liquidity and funding liquidity. *Rev. Financ. Stud.*, 2009, **22**, 2201–2238.
- Byrne, J.P., Korobilis, D. and Ribeiro, P., On the sources of uncertainty in exchange rate predictability. *Int. Econ. Rev. (Philadelphia)*, 2018, **59**, 329–357.
- Campbell, J.Y. and Shiller, R.J., The dividend-price ratio and expectations of future dividends and discount factors. *Rev. Financ. Stud.*, 1988, **1**, 195–228.
- Campbell, J.Y. and Thompson, S.B., Predicting excess stock returns out of sample: Can anything beat the historical average? *Rev. Financ. Stud.*, 2008, **21**, 1509–1531.
- Campbell, J.Y. and Yogo, M., Efficient tests of stock return predictability. *J. Financ. Econ.*, 2006, **81**, 27–60.
- Carter, C. and Kohn, R., On gibbs sampling for state-space models. *Biometrika*, 1994, **81**, 541–553.
- Casarin, R., Grassi, S., Ravazzolo, F. and van Dijk, H.K., Parallel sequential Monte Carlo for efficient density combination: The DeCo MATLAB toolbox. *J. Stat. Softw.*, 2015, **68**, 1–30.
- Chan, J., Koop, G. and Potter, S., A new model of trend inflation. *J. Bus. Econ. Statist.*, 2013, **31**, 94–106.
- Chib, S. and Zeng, X., Bayesian strategy for improved forecasts of the equity premium. Working paper, Olin Business School, Washington University, 2016.
- Christiansen, C., Schmeling, M. and Schimpf, A., A comprehensive look at financial volatility prediction by economic variables. *J. Appl. Econom.*, 2012, **27**, 956–977.
- Clark, T.E. and West, K.D., Approximately normal tests for equal predictive accuracy in nested models. *J. Econom.*, 2007, **138**, 291–311.
- Cogley, T. and Sargent, T., Drifts and volatilities: Monetary policies and outcomes in the post WWII US. *Rev. Econ. Dyn.*, 2005, **8**, 262–302.
- Cremers, M.K.J., Stock return predictability: A Bayesian model selection perspective. *Rev. Financ. Stud.*, 2002, **15**, 1223–1249.
- Dangl, T. and Halling, M., Predictive regressions with time-varying coefficients. *J. Financ. Econ.*, 2012, **106**, 157–181.
- De Mol, C., Giannone, D. and Reichlin, L., Forecasting with a large number of predictors: Is bayesian regression a valid alternative to principal components? *J. Econom.*, 2008, **146**, 318–328.
- Dickey, D.A. and Fuller, W.A., Likelihood ratio statistics for autoregressive time series with a unit root. *Econometrica*, 1981, **49**, 1057–1072.
- Diebold, F.X. and Mariano, R.S., Comparing predictive accuracy. *J. Bus. Econ. Statist.*, 1995, **13**, 253–263.
- Doz, C., Giannone, D. and Reichlin, L., A two-step estimator for large approximate dynamic factor models based on Kalman filtering. *J. Econom.*, 2011, **164**, 188–205.
- Driesprong, G., Jacobsen, B. and Maat, B., Striking oil: Another puzzle? *J. Financ. Econ.*, 2008, **89**, 307–327.
- Durbin, J. and Koopman, S.J., *Time Series Analysis by State Space Methods*, 2nd ed., 2012 (Oxford University Press: Oxford).
- Fagin, S.L., Recursive linear regression theory, optimal filter theory, and error analyses of optimal systems. *IEEE Int. Convention Record Part*, 1964, **1**, 216–240.
- Foster, F.D., Smith, T. and Whaley, R.E., Assessing goodness-of-fit of asset pricing models: The distribution of the maximal R^2 . *J. Finance*, 1997, **53**, 591–607.
- Geweke, J., *Contemporary Bayesian Econometrics and Statistics*, 2005 (Wiley: New York).
- Geweke, J. and Whiteman, C., Bayesian forecasting. In *Handbook of Economic Forecasting*, edited by G. Elliott and A. Timmermann, vol. 1, 2006 (Elsevier: Amsterdam).
- Gneiting, T. and Ranjan, R., Comparing density forecasts using threshold- and quantile-weighted scoring rules. *J. Bus. Econ. Statist.*, 2011, **29**, 411–422.
- Goyal, A. and Welch, I., A comprehensive look at the empirical performance of equity premium prediction. *Rev. Financ. Stud.*, 2008, **21**, 1455–1508.
- Groen, J.J.J., Richard, P. and Ravazzolo, F., Real-time inflation forecasting in a changing world. *J. Bus. Econ. Statist.*, 2013, **1**, 29–44.
- Jansson, M. and Moreira, M., Optimal inference in regression models with nearly integrated regressors. *Econometrica*, 2006, **74**, 681–714.
- Jazwinsky, A.W., *Stochastic Processes and Filtering Theory*, 1970 (Academic Press: New York).
- Johannes, M., Korteweg, A. and Polson, N., Sequential learning, predictability, and optimal portfolio returns. *J. Finance*, 2014, **69**, 611–644.
- Jore, A.S., Mitchell, J. and Vahey, S.P., Combining forecast densities from VARs with uncertain instabilities. *J. Appl. Econom.*, 2010, **25**, 621–634.
- Kascha, C. and Ravazzolo, F., Combining inflation density forecasts. *J. Forecast.*, 2010, **29**, 231–250.
- Kilian, L. and Vigfusson, R.J., Do oil prices help forecast U.S. real GDP? The role of nonlinearities and asymmetries. *J. Bus. & Econ. Statist.*, 2013, **31**, 78–93.
- Kim, C.J., Morley, J.C. and Nelson, C.R., The structural break in the equity premium. *J. Bus. Econ. Statist.*, 2005, **23**, 181–191.
- Kim, C.J. and Nelson, C.R., *State Space Models with Regime Switching Classical and Gibbs Sampling Approaches with Applications*, 1999 (MIT Press: Massachusetts).
- Kim, S., Shephard, N. and Chib, S., Stochastic volatility: Likelihood inference and comparison with ARCH models. *Rev. Econ. Stud.*, 1998, **65**, 361–393.
- Koop, G. and Korobilis, D., Forecasting inflation using dynamic model averaging. *Int. Econ. Rev. (Philadelphia)*, 2012, **53**, 867–886.
- Koop, G. and Korobilis, D., Large time-varying parameter VARs. *J. Econom.*, 2013, **177**, 185–198.
- Koop, G. and Korobilis, D., A new index of financial conditions. *Eur. Econ. Rev.*, 2014, **71**, 101–116.
- Koopman, S.J., Lucas, A. and Scharth, M., Predicting time-varying parameters with parameter-driven and observation-driven models. *Rev. Econ. Stat.* 2016, **98**, 97–110.
- Lettau, M. and Van Nieuwerburgh, S., Reconciling the return predictability evidence. *Rev. Financ. Stud.*, 2008, **21**, 1607–1652.
- Lo, A.W. and MacKinlay, A.C., Data-snooping biases in tests of financial asset pricing models. *Rev. Financ. Stud.*, 1990, **3**, 431–467.
- Maheu, J.M., Song, Y. and Yang, Q., Oil price shocks and economic growth: The volatility link. *Int. J. Forecast.*, 2020, **36**, 570–587.
- Meese, R. and Rogoff, K., Empirical exchange rate models of the seventies: Do they fit out of sample? *J. Int. Econ.*, 1983, **14**, 3–24.
- Mele, A., Understanding stock market volatility. *London School of Economics Financial Market Groups Rev.*, 2005, **67**, 10–15.
- Mele, A., Asymmetric stock market volatility and the cyclical behavior of expected returns. *J. Financ. Econ.*, 2007, **86**, 446–478.
- Neely, C.J., Rapach, D.E., Tu, J. and Zhou, G., Forecasting the equity risk premium: The role of technical indicators. *Manage. Sci.*, 2014, **60**, 1772–1791.

- Nelson, L. and Stear, E., The simultaneous on-line estimation of parameters and states in linear systems. *IEEE. Trans. Automat. Contr.*, 1976, **21**, 94–98.
- Nonejad, N., Forecasting aggregate stock market volatility using financial and macroeconomic predictors: Which models forecast best, when and why? *J. Empir. Financ.*, 2017, **42**, 131–154.
- Nonejad, N., Using the conditional volatility channel to improve the accuracy of aggregate equity return predictions. *Empir. Econ.*, 2020. Forthcoming.
- Pan, Z., Pettenuzzo, D. and Wang, Y., Forecasting stock returns: A predictor-constrained approach. *J. Empir. Financ.*, 2020, **55**, 200–217.
- Pastor, L. and Stambaugh, R.F., The equity premium and structural breaks. *J. Finance*, 2001, **56**, 1207–1239.
- Pastor, L. and Stambaugh, R.F., Predictive systems: Living with imperfect predictors. *J. Finance*, 2009, **64**, 1583–1628.
- Paye, B.S., Déja vol: Predictive regressions for aggregate stock market volatility using macroeconomic variables. *J. Financ. Econ.*, 2012, **106**, 527–546.
- Paye, B.S. and Timmermann, A., Instability of return prediction models. *J. Empir. Financ.*, 2006, **13**, 274–315.
- Pettenuzzo, D. and Ravazzolo, F., Optimal portfolio choice under decision-based model combinations. *J. Appl. Econom.*, 2016, **31**, 1312–1332.
- Pettenuzzo, D., Timmermann, A. and Valkanov, R., Forecasting stock returns under economic constraints. *J. Financ. Econ.*, 2014, **114**, 517–553.
- Primiceri, G., Time varying structural vector autoregressions and monetary policy. *Rev. Econ. Stud.*, 2005, **72**, 821–852.
- Rapach, D.E., K., J. and Zhou, G., Out-of-sample equity premium prediction: Combination forecasts and links to the real economy. *Rev. Financ. Stud.*, 2010, **23**, 821–862.
- Rapach, D.E. and Zhou, G., Forecasting stock returns. In *Handbook of Economic Forecasting*, edited by G. Elliott and A. Timmermann, vol. 2, 2013 (Elsevier: Amsterdam).
- Ruiz, E., Quasi-maximum likelihood estimation of stochastic volatility models. *J. Econom.*, 1994, **63**, 289–306.
- Stambaugh, R.F., Predictive regressions. *J. Financ. Econ.*, 1999, **54**, 375–421.
- Steel, M., Model averaging and its use in economics. *J. Econ. Lit.*, 2020, **58**, 644–719.
- Timmermann, A., How learning in financial markets generates excess volatility and predictability in stock prices. *Q. J. Econom.*, 1993, **108**, 1135–1145.
- Timmermann, A., Excess volatility and predictability of stock prices in autoregressive dividend models with learning. *Rev. Econ. Stud.*, 1996, **63**, 523–577.
- Tourus, W., Valkanov, R. and Yan, S., On predicting stock returns with nearly integrated explanatory variables. *J. Bus.*, 2004, **77**, 937–966.
- Veronesi, P., Stock market overreaction to bad news in good times: A rational expectations equilibrium model. *Rev. Financ. Stud.*, 1999, **12**, 975–1007.
- Viceira, L.M., Testing for structural change in the predictability of asset returns. Unpublished working paper, Harvard Business School, 1997.
- West, M. and Harrison, J., *Bayesian Forecasting and Dynamic Models*, 1997 (Springer-Verlag: New York).
- Westerlund, J. and Narayan, P.K., Does the choice of estimator matter when forecasting returns? *J. Banking Finance*, 2012, **36**, 2632–2640.

Appendix

A.1. Implementation of the suggested Kalman filter

Let $\theta_{t|t-1} = E[\theta_t | y_{1:t-1}]$ and $\varphi_{t|t-1} = E[\varphi_t | y_{1:t-1}]$ denote the conditional expectations of θ_t and φ_t given information up to time $t-1$. Note, the above notation means that for a given quantity, one has that $x_{t-1|t-1} = x_{t-1}$, where x_{t-1} can be θ_{t-1} , φ_{t-1} , Σ_{t-1}

or Γ_{t-1} . At the start of time t , I compute $\theta_{t|t-1}$ and $\varphi_{t|t-1}$ as $\theta_{t|t-1} = \theta_{t-1|t-1}$ and $\varphi_{t|t-1} = \varphi_{t-1|t-1}$. Then, following Kim and Nelson (1999), among others, $\Sigma_{t|t-1}$ and $\Gamma_{t|t-1}$ are computed as:

$$\begin{aligned}\Sigma_{t|t-1} &= \Sigma_{t-1|t-1} + W_t, \\ \Gamma_{t|t-1} &= \Gamma_{t-1|t-1} + V_t.\end{aligned}$$

Evidently, using $W_t = (\lambda^{-1} - 1)\Sigma_{t-1|t-1}$ and $V_t = (\kappa^{-1} - 1)\Gamma_{t-1|t-1}$, I have that $\Sigma_{t|t-1} = \lambda^{-1}\Sigma_{t-1|t-1}$ and $\Gamma_{t|t-1} = \kappa^{-1}\Gamma_{t-1|t-1}$. The prediction error, $\eta_{t|t-1}$, and variance, $f_{t|t-1}$, for (2.1) can be easily computed as:

$$\begin{aligned}\eta_{t|t-1} &= y_t - H_t\theta_{t|t-1}, \\ f_{t|t-1} &= H_t\Sigma_{t|t-1}H_t' + \exp(\delta_{t|t-1}X_{t-1} + h_{t|t-1}),\end{aligned}$$

where $H_t = [1, X_{t-1}]$ and $\theta_{t|t-1} = (\alpha_{t|t-1} \parallel \beta_{t|t-1})$. I can then calculate the predictive likelihood contribution at time t as $y_t \sim N(H_t\theta_{t|t-1}, f_{t|t-1})$. Following Kim and Nelson (1999), among others, one can compute $\theta_{t|t}$ and $\Sigma_{t|t}$ as:

$$\begin{aligned}\theta_{t|t} &= \theta_{t|t-1} + \Sigma_{t|t-1}H_t'^{-1}\eta_{t|t-1} \\ \Sigma_{t|t} &= \Sigma_{t|t-1} - \Sigma_{t|t-1}H_t'^{-1}H_t\Sigma_{t|t-1}.\end{aligned}$$

Having computed $\theta_{t|t}$ and $\Sigma_{t|t}$, I now follow the idea of the dual Kalman filter and do as follows: I insert $\theta_{t|t}$ into (2.1). Thereafter, I bring $H_t\theta_{t|t}$ to the left hand side, take squares and then the logarithm. This leads to

$$\log \tilde{e}_t^2 = Z_t\varphi_{t|t-1} + \log \varepsilon_t^2, \quad (A1)$$

where $\tilde{e}_t = y_t - H_t\theta_{t|t}$, $Z_t = [X_{t-1}, 1]$ and $\varphi_{t|t-1} = (\delta_{t|t-1} \parallel h_{t|t-1})$, which evolves according to (2.3). Accordingly, (2.3)–(A1) form a linear state-space model. As illustrated in Ruiz (1994) and Kim *et al.* (1998), among others, the last term in (A1), namely, $\log \varepsilon_t^2$, follows a $\log \chi_1^2$ distribution with one degrees of freedom. In their study, which concerns estimating stochastic volatility models using Gibbs sampling, Kim *et al.* (1998) suggest approximating $\log \varepsilon_t^2$ using a mixture of Normal distributions. However, the suggested estimation method cannot handle this approximation because it requires simulation. Therefore, I follow Ruiz (1994) and approximate $\log \varepsilon_t^2$ using a Gaussian distribution with mean, $\tilde{\mu} = -1.2704$, and variance, $\tilde{\sigma}^2 = 4.93$. Of course, one can criticize the approximation. For instance, by using such an approximation, I can risk losing information on the region of the distribution that corresponds to large negative values of the log-volatility process. A solution to this problem can be to standardize the dependent and independent variables. This way, the sample variance is equal to one for all variables and more importantly negative values are unlikely to arise. This is a possibility for me as well. Another possibility is to multiply the dependent variable (i.e. the monthly return on the S&P 500 index) by hundred. Therefore, besides the approach taken in the main text, I also perform the following: (a): All data are standardized prior to estimation and (b): The return on the S&P 500 index at month t is multiplied by 100. Overall, I deem the approach in the main text as appropriate because (a) and (b) lead to qualitatively similar conclusions as those in the main text. Results for (a) and (b) are available upon request. Finally, $\varphi_{t|t}$ and $\Gamma_{t|t}$ are computed as:

$$\begin{aligned}\varphi_{t|t} &= \varphi_{t|t-1} + \Gamma_{t|t-1}Z_t'^{-1}\tilde{\eta}_{t|t-1} \\ \Gamma_{t|t} &= \Gamma_{t|t-1} - \Gamma_{t|t-1}Z_t'^{-1}Z_t\Gamma_{t|t-1},\end{aligned}$$

where $\tilde{\eta}_{t|t-1} = \log \tilde{e}_t^2 - \tilde{\mu} - Z_t\varphi_{t|t-1}$ and $\tilde{f}_{t|t-1} = Z_t\Gamma_{t|t-1}Z_t' + \tilde{\sigma}^2$.

A.2. Sensitivity checks

In this section, I demonstrate that results in the main text are robust to the choice of the prior hyperparameter values on θ_0 and φ_0 , $p(\tau_j | y_0)$

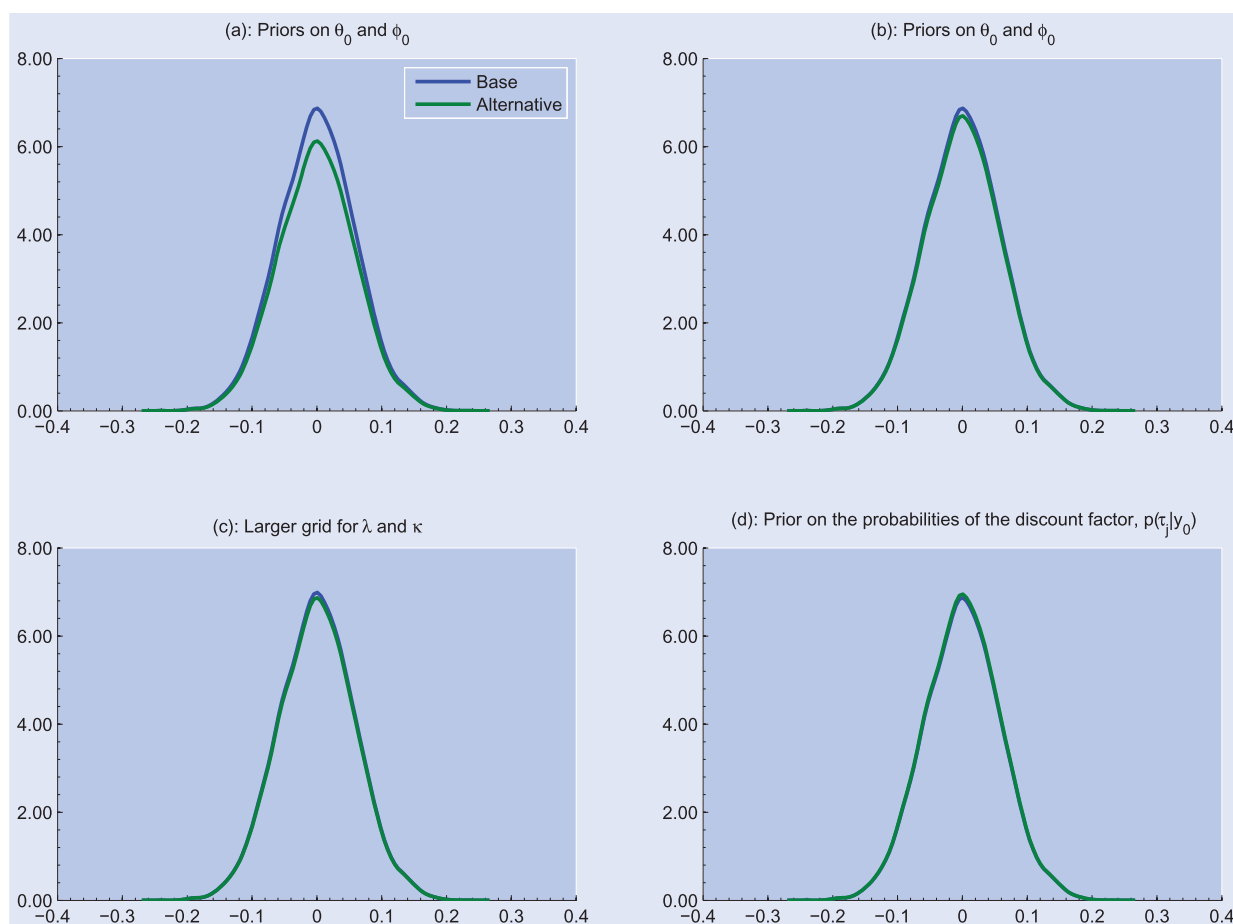


Figure A1. Conditional predictive return distributions for 1947m1 under different scenarios.

These panels display the generated conditional return distributions for 1947m1 given information up to 1946m12 under different scenarios as outlined in Appendix A.2. The model is (2.1)–(2.3), where $X_{t-1} = X_{1,t-1}$ and $X_{1,t-1}$ corresponds to the divided yield. ‘Base’ refers to the model, where input parameters are specified as stated in the main text, whereas ‘Alternative’ refers to the model, where certain input parameters have been changed according to Appendix A.2.

and the grid of discount factor values. I consider (2.1)–(2.3), where $\beta_{1:T} \neq 0$, $\delta_{1:T} \neq 0$ and $X_{t-1} = X_{1,t-1}$ corresponding to the dividend yield. I consider this univariate regression and not BMA because this way, I can better isolate the effects of the mentioned factors and illustrate the consequences without worrying that they could get canceled out by other factors when averaging across different predictor combinations.

To start with, I focus on the prior hyperparameter values on θ_0 and φ_0 . I assume Gaussian priors centered at zero with a very small variance, namely, $\Sigma_0 = \Gamma_0 = 1e - 3I$. This is an extremely tight prior on the initial values of θ_t and φ_t , implying a priori that these vectors are equal to zero. This prior specification is compared to a Gaussian prior centered at zero but with variance equal to $\Sigma_0 = \Gamma_0 = 10I$ identical to the main text. In panel (a) of figure A1, I display the predictive density under each specification for 1947m1 given information up to 1946m12. The latter is labeled as ‘Base’, the former as ‘Alternative’. Accordingly, 1947m1 is the first observation in the out-of-sample period. Thus, any substantial differences between the predictive densities so early on in the out-of-sample period would indicate a degree of sensitivity with regards to the choice of the prior hyperparameter values. However, as displayed in panel (a), although the latter is slightly flatter, both predictive densities are similar even for these very tight (and perhaps unrealistic) prior hyperparameter values on the initial values of θ_t and φ_t . In panel (b), I set $\Sigma_0 = \Gamma_0 = 1e - 2I$, which is slightly looser. Here, results are almost

identical to $\Sigma_0 = \Gamma_0 = 10I$. This is not surprising because simple monitoring of the model shows that it takes at maximum 150 months for the effects of the prior hyperparameters to die out. Given that I work with a long time-series, it does not pose a problem.

Next, I focus on the choice of the discount factor values. In the main text, I consider the combinations of $\lambda \in [0.96, 0.97, \dots, 1]$ and $\kappa \in [0.96, 0.97, \dots, 1]$. For each individual value in $\lambda \in [0.96, 0.97, \dots, 1]$, I consider the grid $\kappa \in [0.96, 0.97, \dots, 1]$. Now, I consider $\lambda \in [0.94, 0.95, 0.96, 0.97, \dots, 1]$ and $\kappa \in [0.94, 0.95, 0.96, 0.97, \dots, 1]$, thereby decreasing the lower bound and increasing the number of combinations. In panel (c) of figure A1, I display the predictive density for 1947m1 under these specifications. However, there are no changes worth mentioning at all. This is expected because I never find that the estimate λ_t or κ_t at any point in time falls below 0.96.

As the final sensitivity check, I consider $p(\tau_j | y_0)$, where $\tau_j = (\lambda_j, \kappa_j)'$ for $j = 1, \dots, d$. In Section 3.1, I assume that $p(\tau_j | y_0) = 1/d$, $j = 1, \dots, d$. Now, I set $p(\tau_1 | y_0) = 0.99$ and then divide 0.01 equally across $p(\tau_2 | y_0), \dots, p(\tau_d | y_0)$. In other words, I a priori assume a very high probability for $p(\tau_1 | y_0) = 0.99$. As indicated by results in panel (d), this does not lead to any changes as both predictive densities are almost identical. Again, this is not surprising as it only takes a small window of observations for the filter to adapt to the data and consequently the probabilities are adjusted according to the dynamics in the data.